Label application: environment report

Environment `label-application` (aliases `label_application`, `apply-label`) · Newton 1.5 rigid world + a dedicated pressure-sensitive-label solver · default robot `dual-fr3` (two FR3 arms, Franka Hands) · action group *Apply label to part* (127 occurrences, 49 leaf groups).

Status. Integrated with GAR (CLI, HTTP, dashboard). Every task is shown solvable by an **assisted** application (labelled placement fixture + phantom applicator at the ASTM D3330 application load) that the unmodified evaluator accepts, including an optical barcode read from an RTX render; predeclared negative controls fail for the stated reasons (§10). **No autonomous or scripted robot solution exists or is claimed:** the scripted robot takes the paper and PET labels off the printer without a tear on all three robot configurations, holds its grasp force and carries the label fingers-down without a tear (§11); its scripted lay and press then fail (the position-servoed press drags the carton while the other hand holds the label, §11). The printer presents the label forward off its peel edge (§7.2). A fingertip bond patch releases under a 5 mm/s peel (§4, §7.5). The paper label's printer-to-carton chain passes the evaluator in one continuous run with a phantom hand taking, carrying and wiping it on (§10.1). The robot-to-helper handoff works: the fingertip patch releases and the robot withdraws cleanly in every run of the final choreography and at every sensitivity setting. The chain after it is not qualified: every run traps more than 20 mm² of air near the label's leading end (§9.1, §10.1, §15 item 16). With the trapped-air solver of §4, 5 of 5 runs stay untorn through the hold and rub. Through the API, with one render worker per operation, light actions run at 22–41× wall/sim with RGB; a contact-heavy take with an RGB image every 0.1 s averages 132–133×, the complete scripted job 144×, and single 0.1 s actions reach 210–458× on physics (§13, §15 item 17). Every number in this report was produced by a run of the environment's code. The limits are listed in §15: read them before relying on the environment.



The parcel pack bench: printer in peel-and-present mode on its stand, 16 x 12 x 8 in carton, tool tray, shipping scale, stock shelving and dock door (RTX render of the initial state).

1. The real job

Scope: what people actually do when they apply a label, and which of those jobs this environment reproduces.

Applying a pressure-sensitive label is a daily step in shipping, clinical specimen handling and receiving. In every setting the sequence is the same: a desktop printer in peel-off mode prints the label and presents it with its trailing edge still on the backing liner; the operator takes it by its free end, carries it to the item, places it where a rule requires, presses it down from one side so no air is trapped, and the barcode must then scan. The rules are external and checkable:

Setting	Label	Rule the placement must satisfy	Source
parcel carrier label	4 x 6 in direct-thermal paper	on the top face, one flap half, not over the tape seam or an edge	carrier packaging guides
GS1 logistic label	4 x 6 in with SSCC GS1-128	on a vertical side, SSCC bottom 32 ± 6 mm above the base, bars vertical, ≥ 19 mm from vertical edges	GS1 Logistic Label Guideline 1.3 [8-4], [8-5], [6-20]
relabel	4 x 6 in	the old label completely covered	carrier relabel practice
specimen tube	2 x 1 in paper	top aligned with the manufacturer label, straight, a viewing window left, clear of the tube bottom	CLSI AUTO12 / reference-lab instructions
part ID	2 x 4 in PET (3M 7818 class)	on the matching marked plate without covering its marking, or horizontal near mid-height on a round HDPE container	receiving / asset-tag practice

Three bench presets share robots, label physics, printer mechanism, cameras and evaluator:

Preset	Setting	Label	Item	Tasks
`parcel`	shipping pack bench	4 x 6 in direct-thermal paper	16 x 12 x 8 in C-flute RSC, 48 mm PP seam tape, 2.55 kg	`carrier-top`, `gs1-case-side`, `cover-old-label`, `press-down`
`specimen`	clinical accessioning bench	2 x 1 in paper	13 x 100 mm PET K2EDTA tube with manufacturer label, on a two-rail rest	`tube-wrap`
`part`	receiving bench	2 x 4 in matte-silver PET film	three marked stainless plates / round 4 L HDPE canister	`part-id-match`, `part-id-hdpe`

The policy sees the task instruction verbatim (`task.py:INSTRUCTIONS`), e.g. for `carrier-top`:

> You are at a shipping pack bench. The desktop label printer has just printed and presented the shipping label for the sealed carton on the bench. Take the label from the printer and apply it to the carton's top, clear of the sealing tape along the seam and clear of the top's edges.

2. Scene at real scale

Scope: what is in the workcell, how big it is and where each dimension and appearance comes from. Geometry a robot or label can touch is physical (scene.py); everything else is visual only (dressing.py).

2.1 Physical geometry (collision, mass, friction)

Object	Dimensions / mass	Provenance
Packing bench	1.83 x 0.76 m, top 0.91 m, laminate, steel frame	common industrial bench size (authored)
Printer, by bench	parcel: 4-inch direct thermal (ZD421d class), housing ~221 x 222 x 170 mm; specimen: 2-inch direct thermal, smaller housing ~190 x 170 x 150 mm; part: 4-inch **thermal transfer** (ZD421t class: the PET 7818 ID label is a thermal-transfer stock and needs a ribbon; ribbon supply and take-up visible in the media window); all on a 0.28 m stand, exit slot 3 mm below / 4 mm above the web	footprint class of the Zebra ZD421 [S7]; housing sizes and details authored
Presenter geometry	trailing 4–10 mm on the liner; presenter lip 35 mm (4 x 6), 12 mm (2 x 4), none beyond the 6 mm peel bar (2 x 1)	**authored fixture**: no referenced printer documents a lip. It sets how far the presented label overhangs for pickup (≥ 15 mm) and carries part of the label adhesive-down on its silicone release surface (490 of 1782 nodes of the 4 x 6 label), so it shapes how a take loads the label
Carton	406 x 305 x 203 mm, 2.55 kg with contents, μ = 0.4, 48 mm seam tape	16 x 12 x 8 in RSC; mass authored
Specimen tube	13 x 100 mm PET, 10.5 g, lavender cap, manufacturer label 0–240°, 18–75 mm from the cap end	BD Vacutainer class dimensions (search); mass estimate
Tube rest	two POM rails 4 mm inside the tube's contact line	authored at the computed contact geometry
Steel bracket	160 x 120 x 6 mm stainless plate	authored
HDPE canister	Ø150 x 250 mm round container with screw lid, 3.2 kg with contents	authored (4 L class)
Tool tray + PP squeegee card	100 x 70 x 1.5 mm card on edge ribs	authored
Robots	two FR3 + Franka Hand on steel pedestals behind the bench, D405-envelope wrist cameras (also `dual-panda`; `single-fr3` on the left pedestal beside the printer)	shared catalog `robots/rigs/label-*.json`

2.2 Visual dressing (no physics)

The room, lights and props are authored into the render USD only (`dressing.py`), never into the physics model: painted block walls, sealed concrete floor and a 600 mm suspended ceiling derived from pinned CC0 Poly Haven scans (`hangar_concrete_floor`, `concrete_block_wall`, `beige_wall_002`; paint and sealer colours applied to the scans' luminance detail, stated in `dressing.py`), four fluorescent fixtures with the rect lights that light the room, steel stock shelving with cartons and totes, a roller-shutter dock door, hand truck, waste bin, fire extinguisher and call point, wall clock and security camera (15 pinned Poly Haven models, CC0; SHA-256 of every file in `visual_assets.json`, downloads verified on every export). The clinical preset stocks its shelving with lab consumables instead of shipping cartons: printed nitrile-glove boxes (M/L), tube-pack cartons, 1 L reagent bottles and a specimen transport cooler. The printer stand carries edge trims, levelling feet and a lower shelf with two spare label rolls. The printer is a detailed two-tone clamshell model with a smoked media window, keypad and status LEDs laid over its collision boxes; the carton carries flexo box markings (the middle third left clear where labels go); the specimen tube is drawn as clear PET with its manufacturer label exactly over the collider's label patch and its 6.0 mL of blood pooled along the lying tube under a horizontal free surface (a circular-segment prism sized from the draw volume and the interior). The overhead policy camera hangs from a ceiling drop rod over the bench; on the specimen bench it uses a narrower lens (28 mm focal on the 36 mm sensor) aimed between printer and tube cradle, which doubles the 13 mm tube's size in the image.

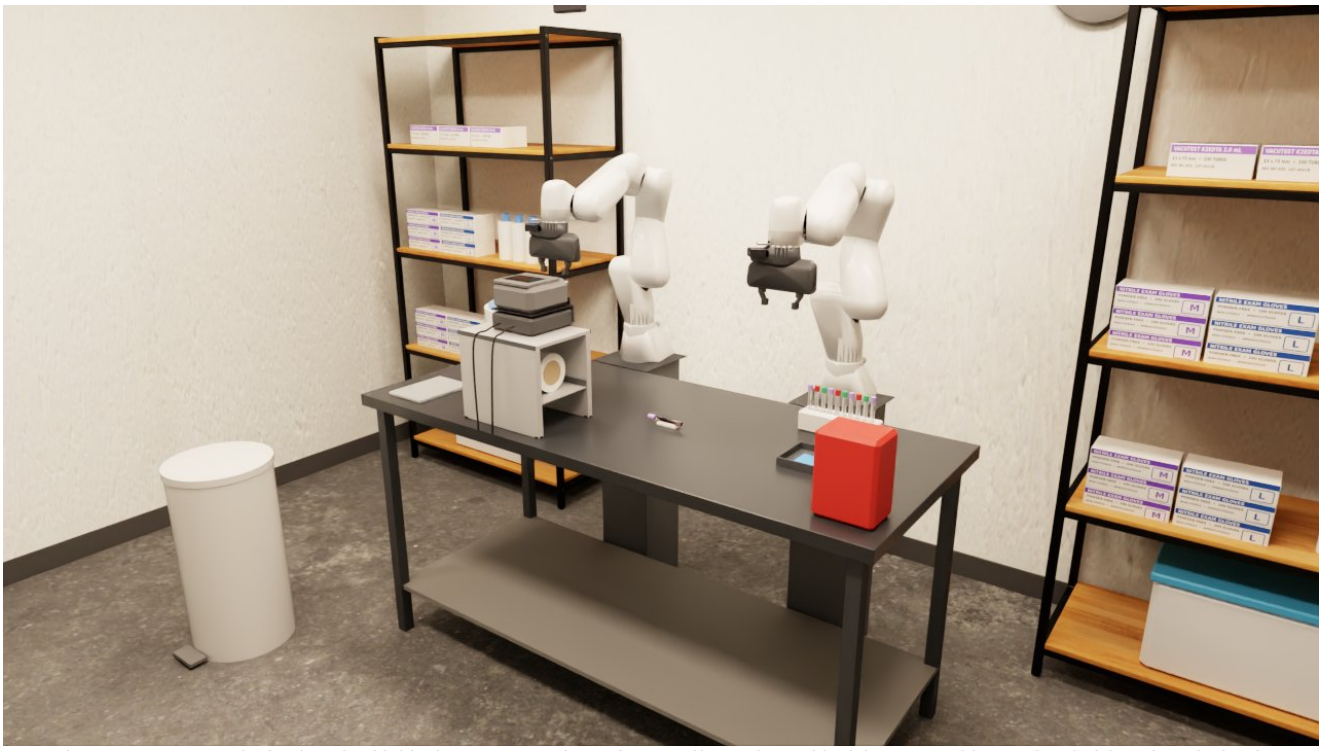
The tube, stock pack and patient label identify a 13 x 100 mm, 6 mL lavender K2EDTA tube, using BD 367863 as the reference class; the authored brand and artwork are not a replica, and the visual blood fill agrees. The specimen images below show an earlier tube artwork; geometry, the estimated 10.5 g mass, contact models, task instructions and scoring are the same.



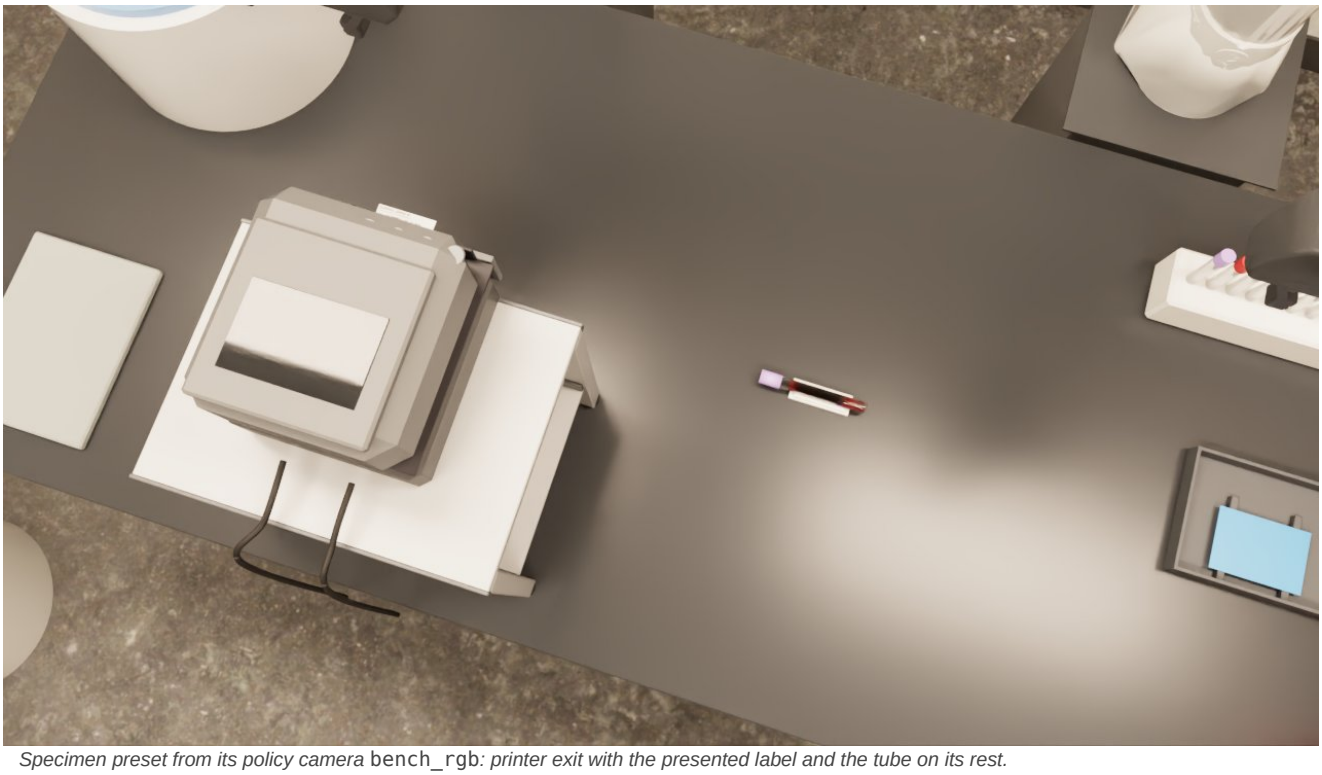
Parcel bench from the overhead policy camera `bench_rgb` *(what a policy sees): printer with the presented label, carton with seam tape, tool tray, scale.*



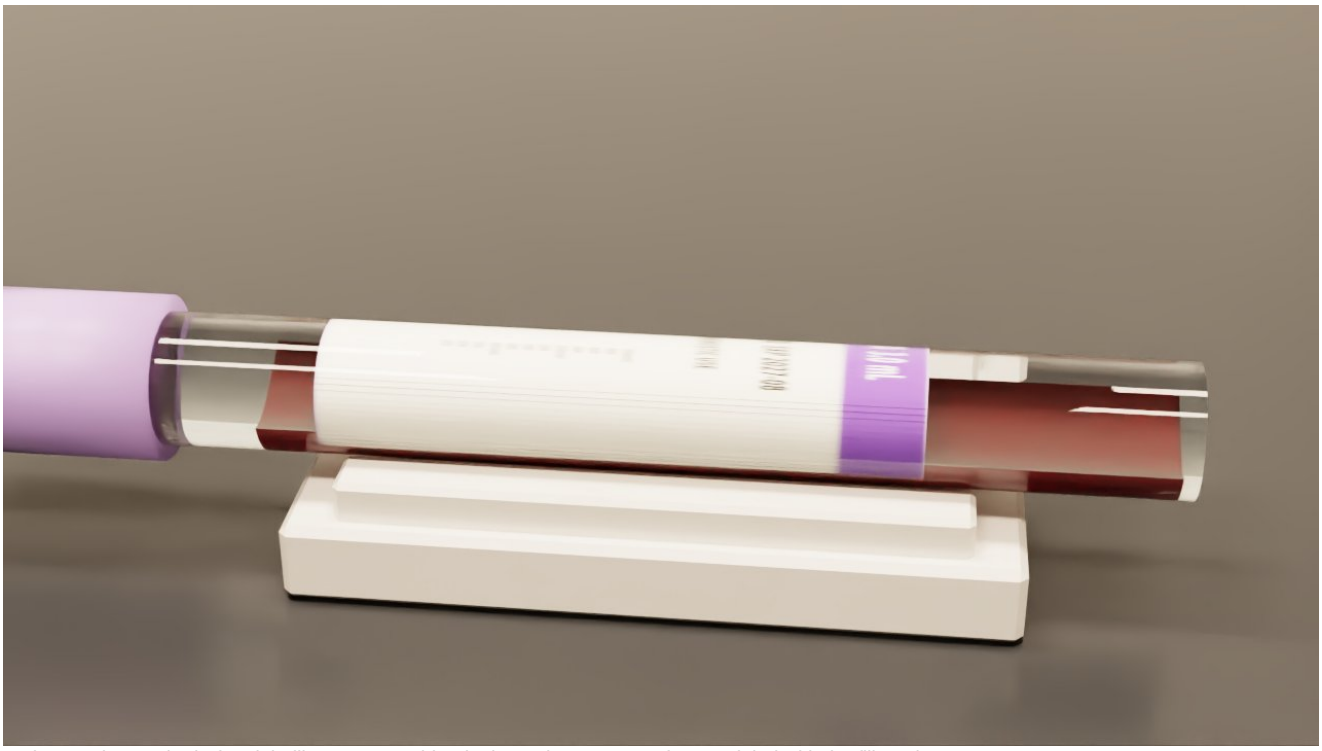
The label printer in peel-and-present mode: 4 x 6 label over the peel bar, spent liner leaving below it.



Specimen preset: accessioning bench with black epoxy top, tube racks, centrifuge, sharps bin, lab consumables on the shelving; the 2-inch printer presents a 2 x 1 label.



Specimen preset from its policy camera `bench_rgb`*: printer exit with the presented label and the tube on its rest.*



The specimen tube before labelling: K2EDTA blood tube on its rest, manufacturer label with the fill mark.

3. Label mechanics

Scope: the solver that decides what the label does. Equations and constants; provenance tags in §6.

The label is one triangle mesh (3 mm for 4 x 6 and 2 x 4 labels, 1.5 mm for 2 x 1), solved every 2 ms by minimising the implicit-Euler incremental potential

$$E(x) = \tfrac{1}{2}\|x - \check{x}\|^2_M/\Delta t^2 + E_{membrane} + E_{bend} + E_{contact} + E_{bond} + E_{friction} + E_{drag} - f_{ext}\cdot x$$

with Newton's method (block-3 sparse Hessian, MKL Pardiso with one symbolic analysis, backtracking line search, 1 µm step tolerance).

- **Membrane:** constant-strain triangles, orthotropic St Venant–Kirchhoff in Green strain, plane-stress stiffness from E_{MD}, E_{CD}, ν and $G_{12} = \sqrt{E_{MD} E_{CD}}/(2(1+\nu))$.
- **Bending:** discrete hinges $E = k_e (\theta_e - \bar{\theta}_e)^2$, $k_e = c_b D(\varphi_e)|e|^2/(A_1+A_2)$, $D = E t^3/(12(1-\nu^2))$, orientation-interpolated between MD and CD. $c_b = 0.5217$ makes an isometric wrap on a cylinder store $\tfrac{1}{2}DA/R^2$ within 0.9 % (MD and CD); gradients match finite differences to 1.9e-9.
- **Permanent set:** a plastic hinge whose rest angle follows once the curvature exceeds $1.5\,\kappa_y$ ($\kappa_y = 2\varepsilon_y/t$; 1.5 = plastic shape factor of a rectangle). Paper $\varepsilon_y = 0.5$ %, PET 3 %.
- **Tearing:** orthotropic Tsai–Hill index on the membrane force per width in material axes, $(S_{MD}/X_{MD})^2 - S_{MD}\,S_{CD}/X_{MD}^2 + (S_{CD}/X_{CD})^2 + (S_{12}/S)^2$, tensile normal components only (paper buckles, it does not crush), $S = \sqrt{X_{MD}\,X_{CD}}/\sqrt{3}$. The stress entering it is the Whitney–Nuismer average over a 2 mm characteristic length (area-weighted mean of the triangles within 2 mm; a single-element peak is a discretisation artefact at a pad corner). A triangle reaching 1 fails: it stops carrying membrane and bending load (element erosion), its edges become free boundary (they vent pockets and can peel like label edges), and a vertex whose every triangle has failed is torn-off debris, frozen where it tore and excluded from contact. A torn sheet therefore separates instead of transmitting intact-sheet loads.
- **Self-contact:** every vertex against every triangle whose rest position is more than 1.5 elements away and farther than the current distance (the sheet has come back onto itself), found by a KD-tree with a 2 mm Verlet skin; unilateral penalty at one label thickness along the lagged separation normal, frictionless and non-adhesive. A folded label cannot pass through itself (§7.6); this also makes a 180° peel meaningful.
- **Air drag:** linear, normal to the sheet.

4. The adhesive interface

Scope: how a label sticks, how firmly, and how it lets go. This is where the task's difficulty lives.

Forming the bond. A vertex whose adhesive face comes within 50 µm of a surface bonds to that surface's material point and can no longer slide (adhesive-layer springs, normal 1.5e10 Pa/m, shear 5e9 Pa/m). Its contact fraction is a running maximum of

$$\varphi = 1 - \exp(-(\bar{p} + p_{tack})/p_r),$$

$\bar{p}$ being the local contact pressure (penalty or rigid sandwich pressure, §5) SUSTAINED over $\tau_w = 1/(2\pi \cdot 1\ \text{Hz}) = 0.16$ s (first-order filter, $d\bar{p}/dt = (p - \bar{p})/\tau_w$, so a single impact does not count as a full application). τ_w is derived from the reference p_r rests on (the adhesive modulus at 1 Hz, the Dahlquist contact time of ~1 s, §6), not measured. Consequences, measured: a pad tapping kraft for one 2 ms step leaves $\varphi \approx 0.04$ where the instantaneous law gave 0.71; the 25 mm hand roller at 20 N wets kraft to 97 % at 20 mm/s, 86 % at 50 mm/s, 68 % at 100 mm/s and 46 % at 200 mm/s; smooth steel saturates at every speed (so the steel calibration is unchanged). $p_{tack} = 1$ kPa quick-stick and p_r the surface's roughness pressure (2 kPa smooth plastics/metal/glass, 20 kPa paper facestock, 40 kPa corrugated kraft). φ scales the bond's fracture energy: a label laid on kraft sticks by tack but peels far more easily than a pressed one.

Breaking the bond. At every bond-front node (bonded, with an unbonded neighbour) the energy released by advancing the front is the Rivlin–Kendall balance of the free strip behind it,

$$G = c_E(h)\,[T\,(1 - \cos\theta) + T^2/(2\,E\,t)],$$

T the strip's membrane force per width along the strip direction (the mean of the free neighbours) and θ its angle to the substrate, both read in the free strip outside the peel's bending boundary layer (3ℓ from the front, $\ell = \sqrt{(D/T)}$; clipped to 2–5 elements). The first term is the load's work (peel), the second the strip's stored stretch — at 0° it is the whole of it, which is Kendall's lap-shear (mode II) limit $F/b = \sqrt{(2Et\Gamma)}$; at 90° $G = T$.

Narrow bond patches. The strip reading assumes a uniform strip behind the front. At a bond patch narrower than the sheet around it — a fingertip pad (17.5 × 18.5 mm) inside the 101.6 mm parcel label — it fails in two ways, traced per node. The patch corners read the peeled arm mixed with the unloaded paper lying flat beside the pad (4–550 J/m² where the arm pulled with 1.6–2.7 kN/m at ~90°). And once a row released, it lay flat on the pad (the fold is sub-element) while the load funnelled into the patch within a few mm: the strip read 0–60 N/m at ~0° where the patch front's bonds carried 100–700 N/m almost normal to the pad. The front held and the paper tore at a corner before the patch let go. So a patch whose front is narrower than the sheet also reads Kendall's balance as a whole (the patch balance). The pull the rest of the label exerts on the patch is minus every external force on it: its bond forces, and all contact forces on it and its 1-ring, which hold the fold's fulcrum. That pull is divided by the length of the patch's loaded front line, at the pull's angle to the substrate, with the same $c_E(\theta)$. The patch's front-line nodes take the larger of the two values. It applies only where the patch itself, measured across the pull, is narrower than 80 % of the sheet's width at its front (PATCH_NARROW). A front as wide as the sheet is a uniform strip, which the strip reading describes and the calibration was built on. (Measuring the loaded front line instead fails on a wide carton bond with an oblique front under the roller and the hand's tack pull: only ~20 mm of front carries half the peak bond force, the wide bond is read as a fingertip patch with its whole pull on 20 mm, G rises from 0.3 to 40 J/m², rows the roller laid let go, and an assisted wipe-on traps 357–2417 mm² of air where this rule leaves 6 mm².) The calibrated 90° peel reads 386.6 N/m with the patch balance (386.7 N/m in the calibration). Two extensions were built for a patch's last row, where the strip reading has nothing to read: Kendall's balance for an island with no interior node (one re-tacked row), and Kendall's 0° term $T^2/2Et$ for a patch pressed onto its substrate but pulled in its plane. They release the pinch probe's re-tacked tab and the robot handoff's last fingertip nodes, and leave the peel checks unchanged, but every version (ungated; gated to patches no presser on another body touches) made the assisted wipe-on trap air or tear in its rub. They are kept switchable and **off**; without them the pinch probe's re-tacked island does not release and the strip tears (§7.5). The handoff's withdrawal re-tack is the robot's retreat pushing its pad into the label: with a clearance-seeking withdrawal nothing re-tacks (§10.1, §15 item 16). Two other rules fail the independent checks. Reading every free-neighbour direction in its own one-element corridor and taking the largest G biases low loads: the 0.5 mm/s peel reads 25 % low. Reading the front node's bond force as its line load puts the calibrated peel at 219 N/m: near a sub-element fold that force includes the lever that presses the released row onto the substrate.

Bending release. At label edges and fronts the bending estimate $G = m^2/(2D)$ (flagging on radius R: $D/(2R^2)$) applies when larger. It is read in three places. The first is the hinges on the front row. The second is the first fully bonded hinge behind the front, or behind a label edge, whose own row has no hinge parallel to it. The third is the curvature κ_s of the surface under the front node's bond, $D\,\kappa_s^2/2$ capped by the facestock's plastic moment. A bonded strip forced to curve — a label following its liner round a peel bar — is held to the surface's curvature right up to the front: the stiff adhesive's process zone, $(4D/k)^{\wedge}¼ \approx 0.5$ mm, is shorter than an element. The springback it stores is released as the front advances. Reading only the front row's hinges misses it: the leading edge never feels its bend, and later rows are carried round the peel edge past yield before they let go, so a dispensed label curls (§7.2). On flat substrates the two added terms are identically zero.

The front moves by the Gent–Schultz / Maugis–Barquins kinetic law inverted for speed,

$v = v_{ref}\,((G/(\varphi\, g_{dwell}) - w_0)/(\Gamma_{90} - w_0))^{(1/n)}$, $g_{dwell} = 1 + \alpha \log_{10}(t_{bond}/600\ s)$,

with Γ_{90} the 90° peel energy of the surface (3M 7818 datasheet: steel 380, PP 180, glass 370, HDPE 180, LDPE 130 J/m²), α from the datasheet's 10 min → 72 h ratios, $w_0 = 0.1$ J/m², $n = 0.6$ and v_{ref} the datasheet test speed (5 mm/s peel, 127 mm/s TLMI liner release). The front's progress through a node is its damage d ($\int v\,dt$ / spacing); the node's bond stiffness is scaled by $(1 - d)$ — the part of its area still bonded — and the node releases at $d = 1$. A released vertex cannot re-bond while it is still in contact (an interface that just failed under shear would otherwise heal every step and a lap joint could never slide off); it may bond again after it separates (re-application).

$c_E(h)$ is the only numerical calibration: a node row releases as a whole and frees $h(1 - \cos\theta)$ of strip at once, which a continuum peel never does, so the mean peel force under-reads Γ by an amount that vanishes as $h \to 0$ (0.65 / 0.88 / 0.94 of the target at 3 / 2 / 1.5 mm, uncalibrated). $c_{90} = 0.664$ / 0.9025 / 0.904 is fitted at ONE point, 90° PET on steel at 5 mm/s = 380 N/m, and $c_{180} = 0.76$ / 0.844 / 1.93 against the elastic Kendall solution at 180° (§7.1, §8). Every other angle, substrate and speed is a check. The patch balance and the bending-release terms do not act on the calibration peels.

Trapped air. Unbonded regions are found each step; a region touching the label edge vents, an enclosed one is an isothermal ideal gas of fixed amount n. Its energy $-n \ln V + p_{atm} V$, with the pocket volume linear in the label's normal motion over the step, enters the implicit step exactly: the rank-one Hessian $n/V^2\ a\, a^T$ that couples every vertex of the pocket is applied with the Woodbury identity (one extra right-hand side per pocket), and $\ln V$ is a barrier the line search respects. Gas is redistributed at uniform pressure when a pocket changes shape, split or merged. The air pushes the substrate with the force it exerts on the label, reversed, so a rigid body under a pocket feels only the label's weight at rest. (A per-vertex diagonal gas stiffness instead lets a 3744 mm² pocket collapse 280x in one 1 ms step.)

A step with pockets has converged only when, besides the 1 µm displacement test, each pocket's last volume update is below 0.1 % of its volume. Under the log barrier a Newton step can at most about double a small pocket's volume, so for a pocket squeezed to a micrometre-scale gap the displacement test alone was met while the gas pressure was still orders of magnitude out of balance.

Each pocket's volume enters the step measured against its substrate's current pose, at the start positions carried with the substrate since the last label step (as the fingertip pads read contact, §5), and is linear about them; the Newton iterate starts there. The substrate's motion into a pocket therefore compresses its air, and a pocket rides with a moving substrate. Measured at the uncarried positions instead, a substrate that has moved further than the pocket's gap since the last step leaves the pocket's vertices below the gap floor, and the floor-clamped start volume credits their climb back to the surface as air: the gas sucks the sheet into the substrate.

Trapped air keeps the surfaces apart where the sheet stands off. A pocket vertex within the adhesive contact band (the solver's contact criterion for every vertex) is in contact: it may bond, and its share of the pocket's air goes to its unbonded neighbours. So the air gathers into a bubble until the remaining vertices stand beyond the band. A pocket's last vertex keeps its air. A newly enclosed pocket holds atmospheric air in its volume, including the 0.1 µm gap-floor film, so it is sealed: lifting a region enclosed while lying flat opens a vacuum. The film is a regularization, not air: a vertex that bonds takes its film with it, so a pocket's films never gather into a bubble.

Without these rules a thin pocket squeezed by the roller or by a carton springing back stays pinned at the 0.1 µm gap floor: the MPa read is never applied, its vertex is frozen (even inside the substrate), it shakes the carton through the label's reactions, and later releases its stored $n \ln(V/V_{floor})$ as a kinetic kick that can tear the label under the rub.

Printer and liner. The liner and the silicone-coated exit plate (peel bar and presenter lip) are the same release chemistry and use the liner's datasheet-anchored law (Γ = 2 J/m² at 127 mm/s, inside the TLMI range 1.9–19 J/m²). In the dispensing component test the liner is a driven web over a 1 mm peel edge (`LinerWeb`), and separation there comes from the bending-energy release, not from a script.

5. Contact and coupling with the robots

Scope: how pressure from fingers and tools reaches the label, and how the label pushes back on rigid bodies.

- **Rigid world:** Newton 1.5 `ModelBuilder` with the shared rig, `SolverMuJoCo` on the CPU at 1 kHz (MuJoCo-Warp cost 65 ms per tick for this two-arm scene; CPU MuJoCo 0.7 ms). Cylinders use 32-facet hulls for rigid contact (single-point cylinder contact lets tube and canister creep).
- **Label contact:** all vertex–collider pairs within 1.5 mm; penalty stiffness per area of the real interface (adhesive layer 1.5e10 Pa/m on smooth items, 5e9 on kraft/paper, 1e9 on elastomer pads); lagged smooth Coulomb friction on the facestock side. Boxes are rounded (fingertip pads 2 mm, other robot boxes 1 mm) so contact normals are continuous around edges. Pairs and their tangent planes are collected once per label step from the start-of-step positions against the colliders' end-of-step pose, **except by the robot's fingertip pads, which read the label in their own frame**: the start positions carried with the pad since the last label step, so a pad sweeping 1–2.5 mm per step meets the sheet one step late instead of by a phantom penetration equal to its step. Other rules fail, each traced per step:
- no carrying: a label held in a fingertip moving 1.6–2.5 mm per 2 ms step was compared with the pads already moved that far; tangent planes came out 30–50° off with up to 1.4 mm of phantom penetration (5–13 N per node), and the carry turn tore the parcel label;
- every vertex at its predicted position $x_0 + \Delta t \cdot v$: a released trailing edge was extrapolated across the printer's sharp box corners, and 4 of 18 parcel takes tore;
- every vertex carried with every moving collider: the tube finger, sliding round the tube, dragged the free sheet's geometry with it and the wrap tore at placement;
- carrying only the vertices a hand holds (bonded; then also pinched; then also touched in the last step): nodes newly met by the sweeping pad still saw its whole step, and the carry turn ran at 1.6, 1.2 and 0.57 kN/m (median) instead of 0.18;
- and the reference pose must be the one the label used in the last label step, because a held item is re-posed twice per step.
- **Single-pad contact:** where a vertex touches one fingertip pad but not its partner (paper draped over the fingertip end), the pad acts with its true rounded-box geometry rather than the pinching-face rule below.
- **Fingertip pads:** the rubber tip touches a sheet through its pinching face (the face toward the partner pad): a point inside the pad's compression zone is pushed along that face's normal only, and a point that entered through the pinching face keeps it while it stays in the pad's pinching half (contact history). Nearest-face contact flips a sheet squeezed > 2 mm near the pad's side to the side-face normal and pushes paper in-plane at ~4 N per node, which tears it at pinch onset. A new contact counts as pinched only if the pinching face is the point's nearest face (within 0.1 mm): closed fingertips pressing straight down meet the label with their end faces, and otherwise points under the end face near the inner pinching face fall into the pinch layer, are pushed apart sideways along the finger gap and tear the pressed label. Label-to-pad bonds act in series with the rubber's own compliance (normal 1e9, shear 3.3e8 Pa/m: E/3 over 2 mm), not as a rigid anchor.
- **Sandwich pressure:** MuJoCo's normal force between a fingertip pad and each partner body is spread over the label vertices squeezed between exactly those two bodies and drives φ. A robot cannot wet a label out with pressure it does not apply.
- **Assisted tools press through contact** (`applicator.py`). A phantom roller, pressing finger or test-rig pad is a label-solver collider with no rigid body. It follows a prescribed tangential path, and along the surface normal it is force-controlled: each step its depth moves until the contact force the label actually exerts on it equals the commanded load (tangent contact stiffness x contact area; out of contact it approaches, but never deeper than the bare item surface). Wet-out comes from the penalty contact pressure it produces, and the load reaches the item through the label's contact reactions. A tool that declared its load as a pressure on every vertex within 1 mm would wet out a label it does not touch: a `hover` control (path 0.8 mm clear, no load) must fail, and does. The wet-out pressure of a vertex is its largest single-pair contact pressure (a sum would count a sandwiched label twice).
- **Squeezed thinner than the label:** soft rigid contact lets closing fingers overlap by ~1 mm. Where a gripping surface and an opposing surface are closer than the label is thick, each is set back by half the deficit (continuous in the geometry), so the label sits at the mid-plane instead of being crushed between two penalty walls (which blows the solve up).
- **Reactions:** label contact, bond and friction forces act on the rigid bodies one step later, through a 5 ms first-order filter and capped at the label's tear load (strength x width) — the explicit coupling otherwise formed an unstable spring loop between a 30 g finger and the printer. They are **momentum-consistent**: the Newton solve stops at a 1 µm update, and with adhesive-layer stiffness ~1.35e5 N/m per bonded node the residual summed over hundreds of nodes would give 0.8–5.5 N one-step kicks on an item the label merely rests on (27 in 10 s on the HDPE canister). The net residual is the sum of the external/interface gradient (membrane and bending forces are internal), and it is removed from the reported reactions in proportion to each collider's load share; the label's motion is unchanged. With it: 0 kicks, same iteration count (a 10 nm tolerance also removes them but never converges during an active peel).

- **Grasp:** the GAR `gripper.inward_effort_n` is a Franka Hand `grasp`: the fingers close at the hand's datasheet speed, 50 mm/s per finger (from the hand's catalog spec; the shared FR3 control profile allows 0.2 m/s), and once blocked (lagging the slewed target by > 0.5 mm) or at the end of their travel the grasp **latches** into a force-controlled hold: the position servo stops pulling (its target follows the finger, leaving its damping) and each finger pushes with exactly the commanded force until the next finger command. A law that scales the squeeze by finger speed (off above 20 mm/s) is a relay that chatters: pinch loads fall to 0 N in 12–68 % of samples and a held label never comes to rest. The fingertip pads' MuJoCo contact stiffness is anchored to the pad compliance the label solver uses (1e9 Pa/m over the 17.5 x 18.5 mm face, two pads in series ≈ 1.6e5 N/m → contact time constant 5 ms, 20 ms elsewhere), and the finger joints' 0 mm lower limit is moved 1 mm lower so the pad contact, not a joint-limit spring, stops the fingers: with the 20 ms default the pads overlapped 1.46 mm and the limit spring took ~7 N of a 20 N grasp. Measured: 20.0 N between the pads for a 20 N command in free air, 19.5–19.9 N while holding a label.
- **Bonds stretched beyond any adhesive's reach are released** (10 mm, above 3x the largest element; a 1 mm cap moved the calibrated 90° peel from 386 to 352 N/m because a discrete front node legitimately opens past 1 mm): this removes nearly fully damaged bonds that the front law never released (otherwise a "liner-bonded" node can stay counted through a 150 mm carry, carrying no load).
- **Hardware facts come from the shared robot catalog**: arm joints and gripper joints (model `groups`), the ready pose (`poses.label_application`), the wrist-camera link (`frames.wrist`), the fingertip-pad geometry and finger speed (the Franka Hand attachment's `contact.fingertip_pads`), the mount poses (one table shared by the scene and the IK), and the qualified rigs (`robots.catalog`). The IK tip is checked against the rig's TCP frame at construction. An attachment without fingertip pads is rejected with that reason; the environment has its own controller profile (`label_application`).
- **Velocity reflex:** a joint faster than 1.05x its datasheet limit stops its arm (targets frozen at the current pose), as the robot's own velocity-violation reflex does; reported in the step info. The joint targets slew at the limits, but the shared servo can overshoot them by up to 1.5x on full-range step targets (measured).

6. Parameter ledger

Scope: every constant that changes an outcome, with where it comes from. D = datasheet, M = measured in this repo, R = derived, C = calibrated here, E = engineering estimate, A = authored scene choice.

Quantity	Value	Tag	Source / reason
Paper facestock thickness, areal mass	83.8 µm, 93 g/m² incl. adhesive	D, R	LabelsOnTime DT spec
Paper E_MD / E_CD	5.0 / 2.5 GPa	E	bounded below by the datasheet strength (≥ 3.4 GPa secant)
Paper tensile strength MD / CD	5.08 / 2.5 kN/m	D / E	datasheet MD; CD at half MD (range 0.37–0.67 MD in the sensitivity study, §9.1)
PET 7818 facestock / adhesive	80 / 20 µm, matte silver	D	3M 7818 TDS, May 2017 revision (pinned; the current sheet has a shorter peel table)
PET E	4.2 GPa	D	Mylar A, ASTM D882
Γ_{90} steel, PP, glass, HDPE, LDPE	380, 180, 370, 180, 130 J/m²	D, R	3M 7818 90° peel table
Γ_{90} kraft, paper facestock, PET tube	250, 250, 320 J/m²	E	no datasheet for the pair
Dwell slope α	SS 0.070, PC 0.066, PP 0.169, glass 0.144, HDPE 0.084, LDPE 0.321 per decade	R	3M 7818 **90°** peel 10 min → 72 h (the model's law is a 90° law)
Liner release	2.0 J/m² at 127 mm/s	D	TLMI release range 1.9–19 J/m²
w_0, n	0.1 J/m², 0.6	E	order from Gent–Schultz literature
Roughness pressure p_r smooth / paper / kraft	2 / 20 / 40 kPa	E, R (range)	least grounded. The smooth-surface value lies inside the range contact mechanics gives: at small load the real contact fraction is $A/A_0 = \kappa\, p/(E^* h')$ (Bush et al.; Persson 2005, Eq. 11, $\kappa = \sqrt{(2\pi)}$), so $p_r = E^* h'/\kappa$; with the adhesive's E* bounded by the Dahlquist criterion ($E' < 0.3$ MPa at 1 Hz → $E^* \approx 0.13$–0.4 MPa) and rms slopes $h' \approx 0.02$–0.1 for finished smooth surfaces (inferred, not sourced), $p_r \approx 1$–16 kPa; the sensitivity study spans 1–20 kPa (§9.1, §15)
Wet-out drive time τ_w	0.16 s	derived (inferred)	$1/(2\pi \cdot 1\ \text{Hz})$: p_r rests on the adhesive modulus at 1 Hz (Dahlquist, the ~1 s contact), so the equilibrium law applies to pressure held ~1 s; a first-order filter reaching 99.8 % at 1 s. No measured PSA wet-out kinetics was found; sensitivity ×⅓..×3 (§9.1)
Bending yield strain paper / PET	0.5 % / 3 %	E	
Front discretisation factors c_{90} / c_{180}	c_{90} 0.664 / 0.9025 / 0.904 and c_{180} 0.76 / 0.844 / 1.93 for 3 / 2 / 1.5 mm meshes, interpolated in $u = 1 - \cos\theta$ (c_{90} at $u = 1$, c_{180} at $u = 2$)	C	c_{90}: 90° PET peel on steel at 5 mm/s on the rate law (380 N/m at 5 mm/s); c_{180}: elastic Kendall at 180° (numerics only); calibrated jointly (§7.1, §8)
Peel plate (silicone exit plate)	2.0 J/m² at 127 mm/s	D	same chemistry as the liner: liner datasheet law
Paper in-plane shear strength	$\sqrt{(X_{MD} X_{CD})}/\sqrt{3} = 2.06$ kN/m	E	Tsai–Hill closure; no datasheet value
Tear characteristic length	2 mm	E	Whitney–Nuismer average-stress length, order of paper fibre length
Fingertip pad compliance (series with bond)	1e9 normal, 3.3e8 Pa/m shear	E	rubber E ≈ 2 MPa over 2 mm; shear E/3
Fingertip rounding, pinch compression zone	2 mm, 2 mm	E	Franka fingertip radius from product photos
Self-contact stiffness, reach, skin	1.5e10 Pa/m, 1.5 mm, 2 mm	A	same stiffness as rigid-surface contact
Standard application pressure	100 kPa	D (web), transfer	3M PSA guidance "approximately 100 kPa (15 psi)" (3M VHB 5952 TDS): a generic firm-application pressure transferred from VHB tape, not a measured law for this label/kraft pair; the peel bench's roller inverts to 88.8 kPa — a check, within 11 % (§9)

Quantity	Value	Tag	Source / reason
Wet-out pass fraction	0.90 of standard	A	task threshold (§9)
ASTM application line load	785 N/m	D	2 kg roller over 25 mm (ASTM D3330)
Peel-bench roller	Ø85 mm, 45 mm wide, 2 kg, applied through contact	D	ASTM D3330 / FINAT roller; a force-controlled phantom tool (§5)
Roller rubber covering	1.6e9 Pa/m	D, R	ASTM roller: ~6 mm of Shore A 80 rubber; E = 9.4 MPa from Gent's Shore relation, over 6 mm (compressible estimate); hand rollers use the same covering
Fingertip-pad rigid contact	MuJoCo time constant 5 ms (20 ms elsewhere); finger lower limit −1 mm	R	anchored to the pad compliance above: two pads in series over the 17.5 x 18.5 mm face ≈ 1.6e5 N/m (§5)
Bond release at extreme stretch	10 mm	A	above 3x the largest element, so it never pre-empts the front law (§5)
Newton step tolerance	1 µm largest update, and each trapped-air pocket's volume update < 0.1 %; reactions made momentum-consistent	A	§4, §5
Trapped-air gap floor	0.1 µm per vertex: a regularization (V > 0), not air; a new pocket holds it as a sealed film at p_atm, a bonding vertex takes its film	A	§4
Fingertip pad adhesion	LDPE proxy, 130 J/m²	E	Franka fingertip polymer not published; swept 2–440 J/m² (§9.1)
Franka Hand force	70 N continuous (35 N per finger)	D	Franka datasheet
Carton 406 x 305 x 203 mm, 2.55 kg, µ 0.4		A	16 x 12 x 8 in RSC
Tube 13 x 100 mm, 10.5 g		E	tube class dimensions

7. Calibration and component validation

Scope: assisted test fixtures (no robot) that check each mechanism against an external reference. Mesh 3 mm unless stated.

7.1 Virtual peel bench (ASTM D3330 / FINAT FTM 1 style) and peel angle

A 25 mm strip gets one 2 kg roller pass at the standard 5 mm/s roll-down (force-controlled through contact), the datasheet's 10 min dwell, and a prescribed test-machine jaw peels it. The jaw moves as a real tester does, relative to the plate at v (1 − cos θ, 0, sin θ) for a front advancing at v (a jaw that follows the discrete front would inject slack). References are evaluated at the measured front speed through the rate law. The Kendall set (PET on PP) loads the strip along the peel direction and then moves the jaw with the steady-peel kinematics of an extensible strip.

Case	Simulated	Reference	Status
PET on steel, 90°, 5 mm/s	386.7 N/m	387.8 N/m (3M 7818 380 at 5 mm/s, rate law at the measured front speed)	calibration point (−0.3 %)
PET on PP, 90°, 5 mm/s	188.9 N/m	184 N/m (3M 7818)	implementation check, +2.7 % (PP's Γ is a model input; not a held-out prediction)
paper on kraft, 90°, 5 mm/s	235.3 N/m	212 N/m (estimate 250 x φ 0.959, rate law)	+11 % (the reference is an estimate; φ 0.959 after the standard 5 mm/s roll-down, 0.894 after a 100 mm/s pass)
PET on steel, 90°, 0.5 mm/s	83.6 N/m	80.3 N/m (rate law at the measured 0.38 mm/s)	+4 %
PET on steel, 90°, 50 mm/s	1609 N/m	1513 N/m (rate law)	+6.3 %
PET on PP, 90° (Kendall set)	184.5 N/m	176.4 N/m	1.046
PET on PP, 30°	1502 N/m	1291 N/m (Kendall, rate-corrected)	+16 %
PET on PP, 0° (lap shear)	12.59 kN/m; front ran at 5.2 mm/s	11.03 kN/m = √(2EtΓ)	+14 % — mode II release exists
PET on steel, 180°, 1.5 / 2 / 3 mm	190.8 / 189.3 / 143.1 N/m	190 N/m, elastic Kendall (the numerical calibration target)	+0.4 / −0.4 / −25 %; 3 mm sits on a bistable response (§15)
PET on steel, 180°, vs the datasheet	as above	470 N/m (3M 7818, 180° 10 min)	the adhesion law is elastic in angle: the datasheet's extra high-angle dissipation (Γ_{180}/Γ_{90} = 2.47 on steel) is bounded, not modelled (§15 item 2)
paper from liner, 180°, 127 mm/s	0.90 N/m	0.97–9.7 N/m (3M release range; elastic 1.0 N/m for the chosen 2 J/m²)	7 % below the range's low end (3 mm 180° branch)

The 180° rows use a 30 mm peel travel (at 3 mm, 15 mm of travel does not reach a steady 5 mm/s front). The numerical factors c_{90} and c_{180} are coupled through their interpolation in (1 − cos θ) and were calibrated jointly (90° within 0.5 % and 180° within 2.6 % of target at 1.5 and 2 mm).

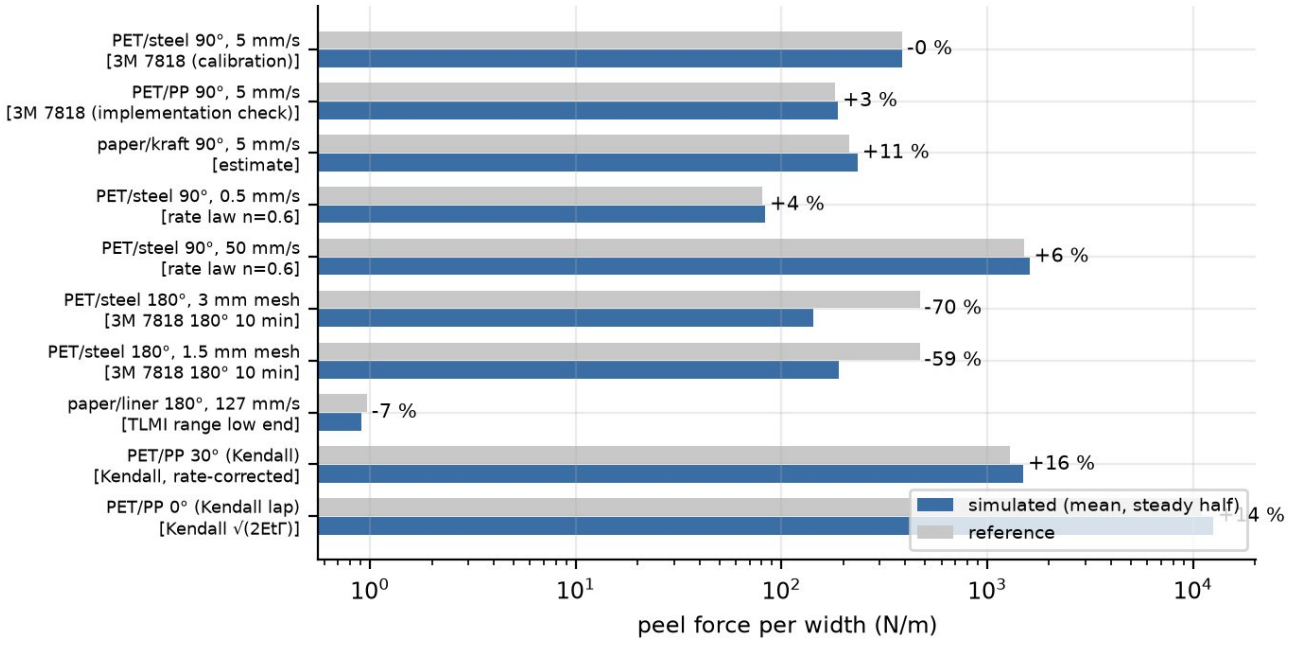


Figure 1. Peel force per width, simulated vs reference (log scale).

The force trace is a sawtooth (each 3 mm row releases in turn, std ≈ 1.4x the mean at 3 mm); the mean matches, the peaks are a mesh artefact. The angle dependence follows Kendall's elastic balance within 16 % from 0° to 90° and, after the 180° numerical calibration, at 180° on the 1.5 and 2 mm meshes. Real acrylic PSAs dissipate more at high angle than an elastic balance holds (2.2–3.4x in Γ on the 7818 substrates); a law for it is not used because the front cannot resolve the peel angle (a true 90° reads a median 69°, a true 180° 119°), so the model under-reads the datasheet at 180° by that factor and the gap is bounded where it matters (§15).

7.2 Dispensing from the liner

Liner web at 80 mm/s over a 1 mm peel edge. Like a peel-mode printer, the web stops at the presentation position, with the label's trailing 10 mm still on the liner (the workcell's `liner_bonded_md`), and then stands still for 1 s. Separation is not scripted: the front advances only where the label's own bending releases more than the liner's rate-dependent release energy (§4). **The paper label presents forward.** It leaves the edge 2° below horizontal and cantilevers straight out, 50 mm past the edge with a 3.4 mm droop at the tip. That droop is what gravity alone gives a 50 mm paper cantilever ($wL^4/8D$ = 2.6 mm, plus the root's compliance). It holds still during the dwell (0.25 mm drift). The result is mesh-converged: at 1 mm the exit angle is 1.9° and the tip sits at (49.87, −3.31) mm; at 2 mm it is 2.0° and (49.87, −3.39) mm. The actual 4 × 6 in parcel label (3 mm mesh) leaves the edge at 12° and hangs forward in a smooth arc under its own weight (152 mm of paper, 142 mm of it past the edge, tip 107 mm below the edge), with no curl. Its trailing 10 mm (140 nodes) stays on the liner, and it is at rest after the dwell. The control label, with 1/100 of the bending stiffness, stays bonded (651 of 651) and follows the liner straight down.

Why the front reads the surface's curvature (§4). A label edge's own row has no hinge parallel to it, so read on its hinges alone the leading edge never feels its bend, and at 1 % yield strain the whole label follows the web down. Once a front forms, each row is then carried round the 1 mm edge until its element bends past yield, and lets go with a permanent set: the label curls into a ~20 mm hook, worse at 1 mm than at 2 mm (a per-step trace shows the front node's G at 0.06–0.68 J/m² until its hinge yields, then 7.2 J/m²). The stiff adhesive holds the strip to the substrate's curvature right up to the front. Its process zone, (4D/k)^¼ ≈ 0.5 mm, is shorter than any element. So the front reads D κ²/2 of the surface under it (the peel arc, 1/r; plastic cap), together with the stored bending of the first bonded hinge behind it. The workcell starts from a presented label over the authored presenter lip (§15 item 13). This component shows that the printer mechanism itself presents the label.

7.3 Trapped air

A cupped PET label on glass: a frame presses the perimeter, then a pad pushes on the enclosed middle, both force-controlled through contact (`applicator.py`; the centre pad may sink at most 3 mm). Edges-first traps one pocket of 1707 mm² at +3.3 kPa overpressure that the pad does not flatten (a position-driven pad instead pushes a 3744 mm² pocket to +12.3 kPa). The same cupped label rolled on from one edge traps nothing: the near edge goes down, a phantom hand holds the far edge up at 15° with a light pull (0.3–1.2 N logged, 0.5 N commanded) and lets go 12 mm from the end, and the bond front follows the 10 N roller across the label (1066 of 1066 nodes bonded, membrane force ≤ 0.13 kN/m). In this probe (sealed air on glass) no pocket goes below atmospheric; pockets on kraft in the robot-to-helper chain do (§15 item 21).

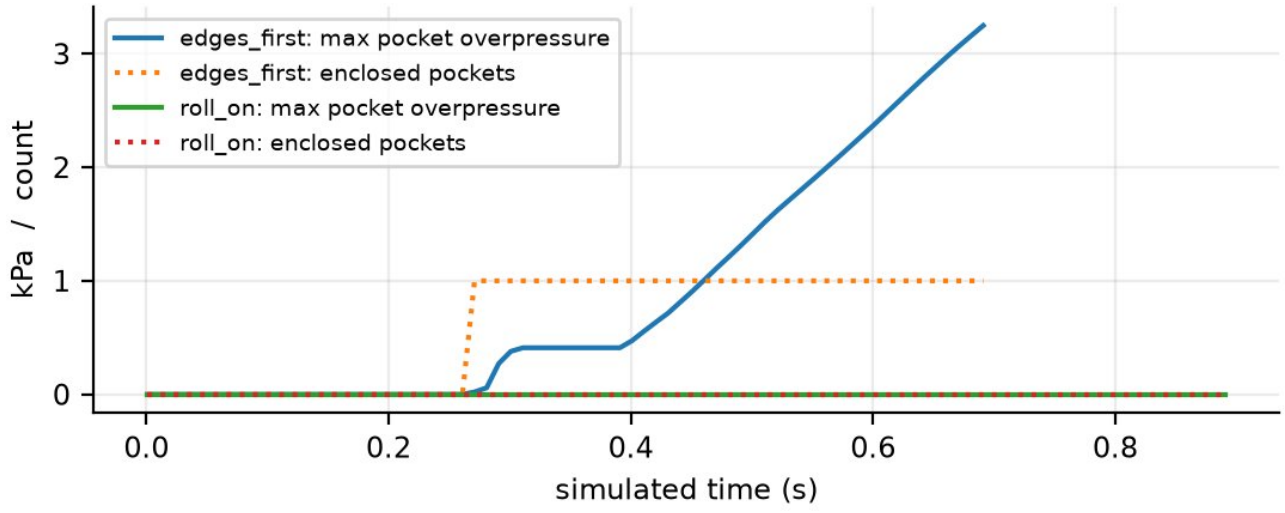


Figure 2. Largest pocket overpressure and number of enclosed pockets, edges-first vs roll-on.

7.4 Wrapping a tube

2 x 1 in label on a 13 mm tube, finger rolled centre-out: pressed edges lift 0.0035 mm (paper) / 0.0028 mm (PET) after 2 s; edges left 3 mm unpressed spring up 0.95 mm (the finger force-controlled through contact at 5

N: 4.98 N measured median) — the flagging a lab rejects.

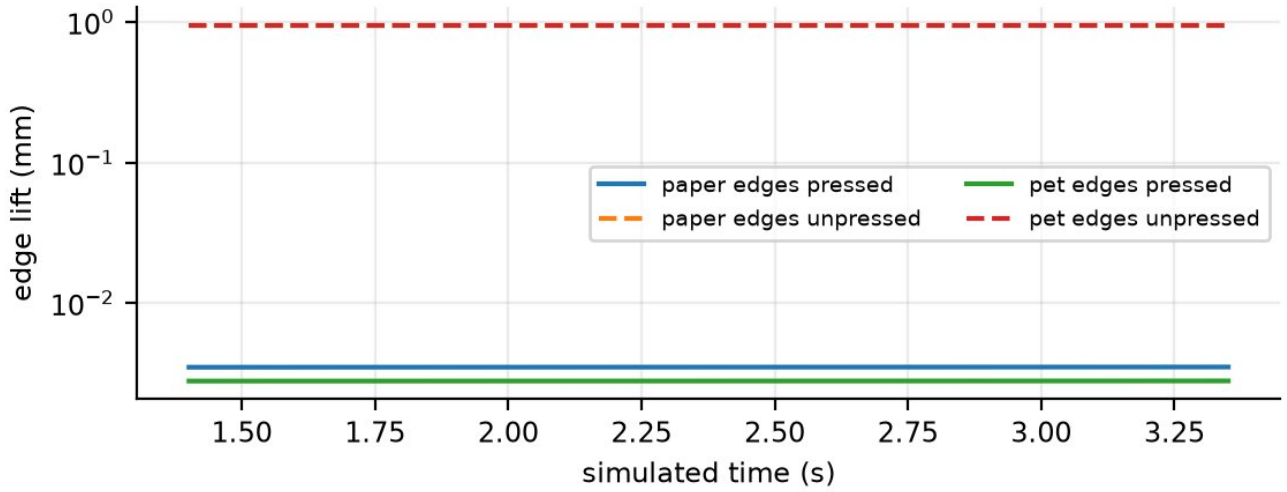


Figure 3. Edge lift after the finger leaves, pressed vs unpressed edges.

7.5 Fingertip pinch (the take, as a component)

Two Franka fingertip pads (URDF geometry, pinch-face contact) close at 20 N on the free end of a 25 x 70 mm paper strip whose trailing 10 mm tab is on a silicone liner, then pull it away and up at 30°, 50 mm/s, for 90 mm. 54–56 nodes bond to the adhesive-side pad, and the tab peels off the liner within ~40 ms of pulling (78 → 52 → 0 nodes by 0.74 s).

- **On the adopted front law the strip tears.** The released tab drops back onto the liner and re-tacks 13 nodes at 0.76 s. That island does not let go: the pull loads it in its own plane, the strip's membrane force rises 0.7 → 2.6 → 4.4 kN/m in 40 ms, and the first triangles fail at 0.812 s (5.4 kN/m; 186 failed by the end).
- **This is the re-tack limitation of §15 item 16, acting in a component test.** The re-tacked island has no interior node and is pulled in its plane while pressed on the liner, and the strip reading returns ~0 there.
- **With the two patch-balance extensions switched on (§4)** the re-tacked rows (26, then 13 nodes) let go within 40 ms. The strip ends released from the liner, untorn: 56 nodes on the pads, peak 2.5 kN/m, Tsai–Hill 0.22.
- **The robot takes themselves are unaffected.** They come off the printer untorn on all three robots and at every sensitivity setting (§9.1, §11). In this probe the tab falls back onto the liner under a 50 mm/s pull, which the demonstrated takes do not do.

Small-patch release: a paper label off one fingertip pad. A paper label still stuck to a fingertip pad is the front at which the scripted lay and the robot-to-helper handoff can tear it (§10.1, §11). This probe isolates that front. A 50 mm wide paper label is bonded to one real Franka fingertip pad (17.5 × 18.5 mm face, 2 mm fillet, pinch-face contact law, LDPE-class proxy, Γ_{90} = 130 J/m² at 5 mm/s), and a test jaw peels it at 90°. The physics has one clear expectation. The line force at the patch front reaches the rate-law Γ(v): 130 J/m² at 5 mm/s, 517 J/m² at 50 mm/s. The paper's strength (5.08 kN/m MD, 2.5 kN/m CD) is 10–40 times higher, so the label must come off, not tear. The long and short full-width strips are the controls. "Adopted" is the front law of §4 (calibrated mean-direction reading, patch balance for patches narrower than the sheet, bending release); "no patch balance" is a front law without the patch balance. The jaw stops 6 mm past the patch; the few nodes still bonded then are the trailing row.

Case	Front law	Bonded, start → end	Torn	Jaw work per area released (Γ)	Peak membrane
long strip, full width, 5 mm/s	adopted	378 → 7	no	178 J/m² (130)	0.76 kN/m
long strip, full width, 50 mm/s	adopted	378 → 15	no	755 J/m² (517)	1.85 kN/m
short strip, full width, 5 mm/s	adopted	108 → 8	no	192 J/m² (130)	0.75 kN/m
short strip, full width, 50 mm/s	adopted	108 → 8	no	823 J/m² (517)	1.84 kN/m
fingertip pad, 5 mm/s	no patch balance	30 → 5	no	1258 J/m² (130)	9.2 kN/m
fingertip pad, 5 mm/s	adopted	30 → 4	no	260 J/m² (130)	3.8 kN/m
fingertip pad, 50 mm/s	no patch balance	30 → 17	**yes**	—	6.7 kN/m
fingertip pad, 50 mm/s	adopted	30 → 3	no	861 J/m² (517)	4.1 kN/m
pad pulled straight off, label held 15 mm away, 5 mm/s	no patch balance	30 → 0	no	—	5.3 kN/m
same	adopted	30 → 0	no	—	3.3 kN/m
same, 50 mm/s	no patch balance	30 → 30	**yes**	—	5.4 kN/m
same, 50 mm/s	adopted	30 → 30	**yes**	—	5.3 kN/m

Case	Front law	Bonded, start → end	Torn	Jaw work per area released (Γ)	Peak membrane
label peeled back at 180°, 5 mm/s	adopted	30 → 4	no	483 J/m² (130)	3.1 kN/m
label peeled back at 180°, 50 mm/s	adopted	30 → 30	**yes**	—	3.3 kN/m

Without the patch balance the patch corners read the peeled arm averaged with the unloaded paper lying flat beside the pad (4–550 J/m² where the arm pulled with 1.6–2.7 kN/m at ~90°), so they held, the front went U-shaped, the paper carried up to 9.2 kN/m at 5 mm/s and tore at 50 mm/s. With the patch balance (§4) the fingertip patch releases untorn at both speeds, and the straight pull releases with 3.3 kN/m instead of 5.3. What is not physical is stated. At 5 mm/s the jaw's work per released area on the fingertip is 260 J/m², twice Γ; on the full-width strips it is 178–192 J/m², so about 50–80 J/m² of the excess is the plastic fold the front drags along, and the rest is the patch. The 180° peel-back needs 483 J/m²: at a 180° front the fold (~1.5 mm) is shorter than the 3 mm element, so just-released nodes touch the pad again and re-bond before they let go. Fast peels of the pad patch on the 101.6 mm parcel label tear (a fingertip opening at ~85 mm/s during the scripted lay), and at 50 mm/s the straight pull and the 180° peel-back tear without releasing a node; the demonstrations peel the patch at 5 mm/s (§15 item 16).

7.6 Self-contact

A PET strip bonded on a plate with its free half folded back on an arc and pressed (force-controlled, 2 N): with self-contact the folded layer rests 0.0988 mm from the lower layer's mid-surface at its closest (label thickness 0.100 mm), 0 of 81 overlying vertices interpenetrate; without it the pad drives 9 vertices more than half a thickness into the lower layer (4 through its mid-surface). The fold does not tear either way.

8. Discretization convergence

Scope: do the numbers the tasks rest on survive mesh and time-step refinement? Gate stated before running: ≤ 5 % change.

Quantity	h = 3 mm, dt = 2 ms (production)	h = 2 mm	h = 1.5 mm	dt = 1 ms	Gate
kraft wet-out φ after the ASTM roller	0.894	0.875 (−2.2 %)	0.895 (+0.1 %)	0.894 (0.0 %)	passes
90° PET/steel peel, calibrated factors per mesh	386.4 N/m	376.0 (−2.7 %)	381.7 (−1.2 %)	386.9 (+0.1 %)	passes; each within 0.5 % of its rate-law target
180° PET/steel peel vs elastic Kendall (190 N/m)	158.1 N/m (−17 %, bistable branch)	191.5 (+0.8 % vs 190; +2.6 % at its measured speed)	190.7 (+0.4 %)	—	3 mm outside the gate, stated (§15)
same 90° peel, uncalibrated (factor 1)	253 N/m (0.65 of target)	346 (0.88)	357 (0.94)	—	the discretisation factor, → 1 as h → 0

The front's discretisation factors are fitted per mesh: c_{90} at the one datasheet point (0.664 / 0.9025 / 0.904 at 3 / 2 / 1.5 mm) and c_{180} against the elastic Kendall solution (0.76 / 0.844 / 1.93), jointly because the two are coupled through the (1 − cos θ) interpolation; the verification runs above then reproduce the 90° point at every mesh and time step. The wet-out the evaluator rests on is converged without any factor.

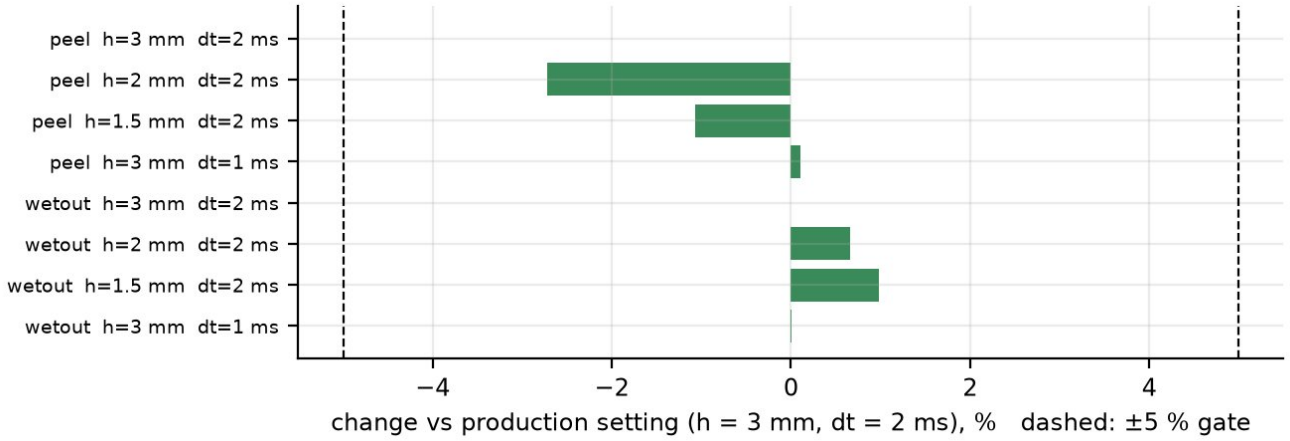


Figure 4. Relative change of the calibration peel force and of the kraft wet-out under refinement.

9. Task evaluation

Scope: what the evaluator measures, why each threshold is where it is, and how the barcode is read.

The evaluator (`task.py`) reads only physical state and renders: it is read-only and never moves anything. Success requires every check:

- **Wet-out ≥ 90 % of a standard application**, over the label area and over its 3 mm edge band. The standard application is 3M's published PSA guidance, "approximately 100 kPa (15 psi)" of roller or platen pressure (web-sourced: 3M VHB 5952 technical data sheet); through the wet-out law it gives reference contact fractions 0.920 (kraft, p_r = 40 kPa), 0.994 (paper facestock) and ≈ 1 (smooth plastics/metal). Independent check: the ASTM D3330 2 kg roller pass in the virtual peel bench left $\varphi = 0.894$ on kraft, which inverts to 88.8 kPa — 11 % below the guidance. **What this reference is and is not:** 100 kPa is a generic firm-application pressure transferred from 3M's VHB tape guidance; it is not a measured pressure-to-contact curve for this paper label on corrugated kraft (none was found), and the 90 % pass fraction is an authored task choice. The contact fractions it implies depend on the estimated roughness pressures, bounded in §9.1.
- edges bonded or lying adhesive-side-down within the 50 µm contact band (up to 3 mm of lifted boundary tolerated); no lifted trapped-air area > 20 mm² (> 3 mm² inside the barcode). Area is integrated over clipped material triangles, not summed whole vertex areas. No permanent set above the barcode curvature threshold; not torn; bonded only to the target; settled — the largest **net** label motion over the last 0.1 s below 1 mm/s (an instantaneous vertex-speed threshold fails 6 of 11 correct tube applications on a zero-mean jitter of ≤ 2.4 mm/s from the lagged coupling on the 10 g tube); item resting on the bench (tube in its cradle), hands and tools off. movement of the item alone is not a failure;
- the task's placement rule (§1): carton labels clear the visible 1.5 mm edge bevel, rather than an unmodeled 5 mm fold radius; tube registration permits 0.5 mm capward / 2 mm below the manufacturer edge, an authored alignment tolerance;
- **optical read at declared reader poses** (`optical.py`): a handheld imager (40° HFOV, 16:9) is held at poses fixed by the item's pose and the task's target zone — never by where the label ended up: over each carton flap half at three positions 0.22 m up (carrier-top), over the old label (0.22 m) (cover-old-label), in front of the long side at two heights and three positions (GS1), a rotating tube reader (fixed imager 90 mm off the tube wall, tube turned in 12 × 30° stops), over the plate (steel), and around the jug in 12 stops (HDPE). Poses are tried in order until the first good read. A read counts only if zxing-cpp's decode equals the printed payload **and** the whole symbol with its quiet zones projects inside that image on the side facing the reader — a clipped symbol that happens to decode is not a verified scan. A reader placed on the label's own centroid and normal is also rendered and reported as `diagnostic_auto_aim`, never used for success.

9.1 Sensitivity of the judgments to the estimated constants

The sensitivity study varies one estimate at a time between a low and a high bound, everything else at baseline, and also sets every estimate at once to its adverse (`coupled low`) or favourable (`coupled high`) bound. It reruns eight outcomes:

- the assisted parcel application (carrier-top), the relabel (cover-old-label), the tube wrap and the HDPE part application, each with the post-application hold and rub (§10);
- the HDPE one-tenth-load control (`hdpe_light`);
- the unassisted robot takes of the parcel and specimen labels;
- the robot-to-helper handoff (the helper's 90° peel off the opened fingertip) up to the robot's retreat (`handoff`).

The applications and takes are run at every setting. The handoff (about 25 min per run) and the two HDPE outcomes are run only at the settings that can act on them.

Setting	parcel	relabel	tube	hdpe	hdpe_light	pickup	pickup_specimen	handoff
baseline	pass (100 %)	pass (100 %)	pass (100 %)	pass (100 %)	pass (99 %)	taken	taken	handed off
kraft_gamma low (125)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	—
kraft_gamma high (500)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	—
p_rough low (1e+04, 2e+04)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	—
p_rough high (4e+04, 8e+04)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	—
paper_cd_strength low (1880)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	—
paper_cd_strength high (3404)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	—

Setting	parcel	relabel	tube	hdpe	hdpe_light	pickup	pickup_specimen	handoff
paper_yield low (0.003)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	—
paper_yield high (0.01)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	—
rate_exponent low (0.4)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	—
rate_exponent high (0.7)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	—
pad_gamma low (2)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	handed off
pad_gamma high (440)	pass (100 %)	pass (100 %)	pass (100 %)	—	—	taken	taken	handed off
p_rough_smooth low (1000)	pass (100 %)	pass (100 %)	pass (100 %)	pass (100 %)	pass (100 %)	taken	taken	—
p_rough_smooth high (2e+04)	pass (100 %)	pass (100 %)	pass (100 %)	pass (99 %)	FAIL (42 %)	taken	taken	—
wetout_time low (0.0531)	pass (100 %)	pass (100 %)	pass (100 %)	pass (100 %)	pass (100 %)	taken	taken	—
wetout_time high (0.477)	FAIL (86 %)	FAIL (95 %)	pass (100 %)	pass (100 %)	FAIL (87 %)	taken	taken	—
coupled low	FAIL (77 %)	FAIL (85 %)	pass (99 %)	FAIL (79 %)	FAIL (23 %)	taken	taken	handed off
coupled high	pass (100 %)	pass (100 %)	pass (100 %)	pass (100 %)	pass (100 %)	taken	taken	handed off

Bounds (baseline in brackets):

- paper-on-paper and paper-on-kraft adhesion Γ ×0.5–×2 [250 J/m²];
- roughness pressure of kraft / paper ×0.5–×2 [40 / 20 kPa];
- paper CD strength 0.37–0.67 of the datasheet MD 5.08 kN/m [2.5 kN/m];
- paper yield strain 0.003–0.010 [0.005];
- Gent–Schultz rate exponent 0.4–0.7 [0.6];
- fingertip-pad adhesion 2–440 J/m² [130 J/m², the 7818 LDPE value at 90°; 440 is its 180° value];
- roughness pressure of the smooth surfaces (HDPE, PET tube, steel) ×0.5–×10 [2 kPa];
- wet-out time constant τ_w ×⅓–×3 [0.16 s, derived].

Application cells give the final verdict after hold and rub, with the wet-out relative to the standard application. Take cells give the result of the unassisted take. Handoff cells give the state after the robot hand has opened and backed away.

- **Applications:** every positive verdict holds across every single bound except one (all 114 results on the front law of §4):
- **the wet-out time constant.** At 3x the derived τ_w (0.48 s) the 20 mm/s roll holds each point too briefly: the parcel application fails (wet-out 86 %) and the relabel fails on its edge band (overall 95 %), against the 90 % thresholds. The kraft verdicts therefore rest on the derived, unmeasured wet-out time constant (§15 item 18); the smooth-surface tasks do not depend on it. The tube passes at twice the paper Γ (without the patch balance the rub tears it there, 41 %) and at the adverse corner (99 %). At the adverse coupled corner (every estimate at its unfavourable bound at once) the parcel (77 %), relabel (85 %) and HDPE (79 %) applications fail; at the favourable corner all pass.
- **HDPE pressure discrimination depends on one estimate.** At the baseline 2 kPa smooth-surface pressure the one-tenth-load control passes (99 %, §10 and §15). At 20 kPa it fails (42 %) while the full-load application still passes (99 %); it also fails at 3x τ_w (87 %) and at the adverse corner (23 %). The environment rejects a light touch on HDPE only if the smooth-surface pressure or the wet-out time is nearer its high bound; no measurement anchors either (§15 items 12, 18).
- **Takes:** the parcel and specimen labels are taken off the printer at every setting and at both corners, with nothing left on the printer and no tear.
- **Handoff** (the helper's 90° peel off the opened fingertip, §10.1; stopped after the robot's withdrawal): the label is handed off untorn, with nothing left on the fingertip, at every setting it was run at. These are the baseline, the weak (2 J/m², where the take leaves only 4–5 nodes on the pad) and the strong (440 J/m²) pad-adhesion bounds, and both coupled corners; the hand withdraws 22–39 mm clear of the label in 1–7 steps. With a choreography whose opening creases the label and whose retreat heads for the label's centroid, all five tear: during the helper's swing at the weak pad bound and at both corners (15.5–15.8 s), during the peel at 440 J/m² (17.49 s, 22 nodes still on the pad), and at baseline at 21.35 s, when the retreat pushes the pad back into the label (35 nodes on the pad). The chain after the handoff (carry, wipe-on, hold, rub) is not part of the sensitivity study; at baseline it applies the label but traps air (§10.1).

10. Assisted end-to-end solvability

Scope: is every task solvable in this world, and does the result survive a hold and handling? Assisted — this is not a policy.

Assistance (in every log line, video caption and output file): (1) a placement fixture lifts the label off the liner and holds it flat, adhesive side down, laid on the rule-compliant position at tack contact (curved items: tangent at the target line); the specimen tube and the HDPE canister are additionally held, as a person holds them, and the carton is braced when its vertical side is rolled (unbraced, the roller's 80 N pushes the 2 kg carton 3 mm across the bench and loses the label); (2) a phantom applicator — a 12.5 mm roller on flat faces and the canister, rolling without slip (`rolling.py`), and a sliding 4 mm finger on the tube — makes one pass at 20 mm/s (flat: from one edge; curved: centre-out). It is **force-controlled through contact** (`applicator.py`): its depth is adjusted each step so that the label's measured contact reaction on it equals the ASTM application line load, 785 N/m times the pressed length; the load reaches the label only by pressing it, and the item only through the label. (3) after a 0.6 s settle, a **2 s hold** with nothing touching the label, then a **handling perturbation**: an 8 mm gloved fingertip (rubber, μ 0.8) slides across the label and back (flat) or along the tube / canister at 5 N and 50 mm/s, then 1 s settle. Phantom tools exist only for the label solver; in the videos they are drawn as translucent amber proxies. Each run's log reports the tool's commanded and measured force, depth and contact area (the carton-top roller: 79.6 N measured for 79.7 N commanded, 136 kPa median contact pressure). The unmodified evaluator judges the result after application, after the hold and after the rub; the barcode is read by the declared-pose reader (§9).

Task	Run	Applied	Hold	Rub	Wet-out / edge (%)	Read	Item moved (mm)	Failed checks
`carrier-top`	hover	fail	—	—	2.7 / 2.7	yes	0.000	wet-out < 90 %, edge band wet-out < 90 %
`carrier-top`	laid	fail	—	—	2.7 / 2.7	yes	0.000	wet-out < 90 %, edge band wet-out < 90 %
`carrier-top`	light	fail	fail	fail	42.8 / 42.9	yes	1.625	wet-out < 90 %, edge band wet-out < 90 %
`carrier-top`	seam	fail	fail	fail	100.0 / 100.0	no	1.332	not over tape seam, one flap half, no read
`carrier-top`	assisted	pass	pass	pass	100.0 / 100.0	yes	1.294	—
`cover-old-label`	assisted	pass	pass	pass	100.0 / 100.0	yes	1.299	—
`gs1-case-side`	assisted	pass	pass	pass	100.0 / 100.0	yes	0.000	—
`part-id-hdpe`	hover	fail	—	—	38.8 / 27.5	yes	0.000	wet-out < 90 %, edge band wet-out < 90 %, lifted edge
`part-id-hdpe`	light	pass	pass	pass	98.9 / 99.0	yes	0.000	—
`part-id-hdpe`	assisted	pass	pass	pass	100.0 / 100.0	yes	0.000	—
`part-id-steel`	assisted	pass	pass	pass	100.0 / 100.0	yes	2.659	—
`tube-wrap`	laid	fail	—	—	0.6 / 0.2	yes	0.026	wet-out < 90 %, edge band wet-out < 90 %, lifted edge
`tube-wrap`	assisted	pass	pass	pass	100.0 / 100.0	yes	0.038	—

All six tasks pass after application, after the 2 s hold and after the rub, with the barcode read at a declared pose. The controls fail where they should: a label only laid on (tack contact, no pressure), a roller held 0.8 mm clear of the label with no load (**hover**, the control for a tool that declares its load instead of pressing: 2.7 % wet-out on kraft), and a light application on kraft (42.8 %) fail wet-out; a label across the tape seam fails placement and is not read from the flap-half poses. On the HDPE canister the hover control fails (38.8 %, tack alone on a smooth surface) but the light-load control **passes** (98.9 %): the estimated smooth-surface roughness pressure (2 kPa) is reached even by light contact held for the roller's contact time. This is reported as a finding about that estimate (§15 item 12), not tuned away.

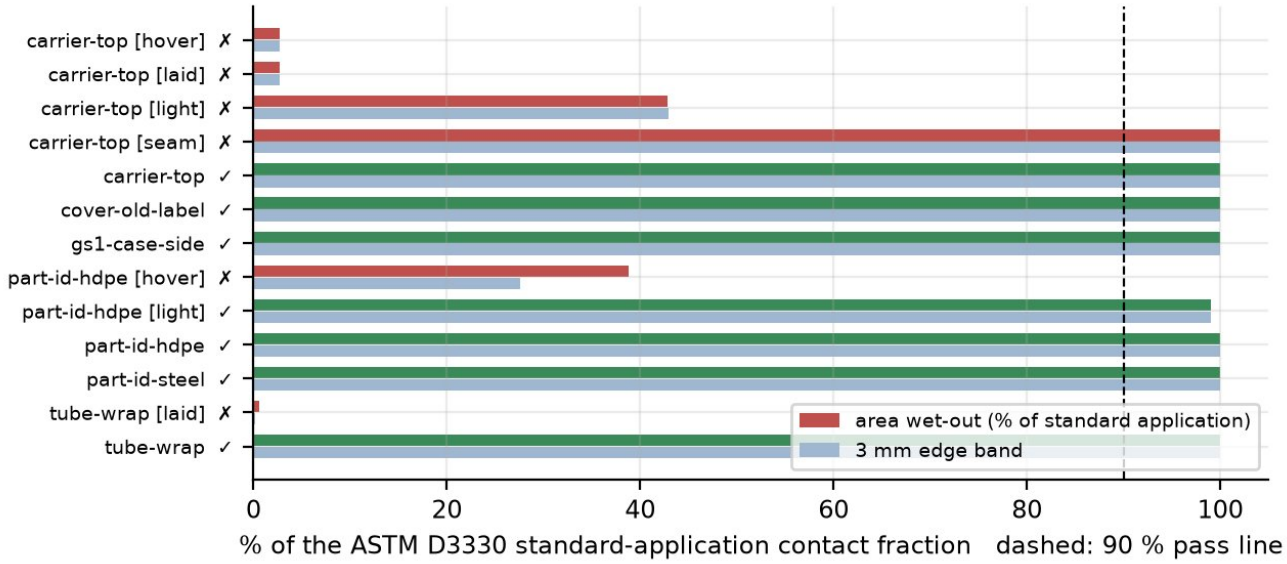


Figure 5. Wet-out relative to the standard application for every assisted run and control; dashed line: the 90 % pass threshold.

10.1 Paper label: printer → take → transfer → application, one continuous run

One simulation runs the whole chain with a printer-presented **paper** label: no mesh reset and no placement fixture. The label comes off the printer's liner, is carried and laid, and is judged by the unmodified evaluator after the same hold and rub as §10, with the barcode read at the declared poses. It is assisted: a phantom hand does what a second person would do. Two variants start from the same printer-presented state.

- **phantom** (qualified): phase A' is a phantom jaw on the label's free end, following the robot take's path (steep lift, then draw away). It then re-grips the label by its middle and trailing edge.
- **robot** (not qualified): phase A is the unassisted robot take (§11). In phase B the phantom jaws close across the label's middle, on its trailing edge and 15 mm behind the fingertip-pad bond. The robot opens its hand at 5 mm/s per finger *around* the fingertip the label is stuck to: the hand moves opposite to that finger's opening, so the stuck pad stays where it is, and the helper's hand follows whatever motion the pad still makes. The helper then swings the label up to 90° about the pad bond's near edge and peels it off the still fingertip at 90°, as a person takes a sticker off someone's finger: it pulls along the pad normal at 5 mm/s and follows the measured front in the pad's plane, as a 90° peel tester does. The robot withdraws its hand in 5 mm steps, each along whichever of 26 directions leaves the most clearance between the label and every collider of the hand, and lifts once the path is clear. The helper draws the label clear and carries it with both grip rows as one rigid frame.

Both then continue the same way. The jaws carry the label, untensioned and adhesive down, 60 mm over the rear flap half. In phase C a third grip takes the free leading edge at carry height, and the sheet is brought down level to 20 mm. It is then applied in one of three ways (`--lay`):

- **wipe** (default): the leading edge is tacked 0.3 mm over its target with the sheet at 7.5°. The middle and leading-edge jaws let go. The rolling roller (ASTM line load) starts on the tacked edge row and runs to the trailing edge, while the hand holds the rest of the sheet up at the tack angle about the roller's contact line (1 % slack). This is how label applicators and people wipe a label on: the sheet meets the carton only under the roller, and the air leaves ahead of it.
- **hinge:** the three jaws lower the straight sheet about the tacked edge, then the roller makes its pass.
- **arc:** only the trailing edge is brought down, keeping the free part an arc tangent to the carton at the actual contact front, then the roller pass.

Every phase records the tear state, bond counts per collider, the fastest label node and the peak membrane force.

Run	Phase	Torn (failed tri.)	Bonds	Max node speed	Peak membrane	Evaluator
phantom take, wipe-on lay (default)	A': phantom hand took the label off the printer	no	none	0.004 m/s	0.06 kN/m	
phantom take, wipe-on lay (default)	B2: carried, held 60 mm over the flap	no	none	0.002 m/s	0.17 kN/m	

Run	Phase	Torn (failed tri.)	Bonds	Max node speed	Peak membrane	Evaluator
phantom take, wipe-on lay (default)	C0: wiped on (roller pass done, hand released)	no	carton 1815	0.001 m/s	0.08 kN/m	
phantom take, wipe-on lay (default)	C1: applied	no	carton 1815	0.000 m/s	0.08 kN/m	wet-out 99.1 %
phantom take, wipe-on lay (default)	C2: after 2 s hold	no	carton 1815	0.000 m/s	0.08 kN/m	wet-out 99.1 %
phantom take, wipe-on lay (default)	C3: after rub + settle	no	carton 1815	0.000 m/s	0.08 kN/m	wet-out 99.2 %
phantom take, wipe-on lay (default)	**final**					**PASS**; reader: read
phantom take, hinge lay	A': phantom hand took the label off the printer	no	none	0.004 m/s	0.06 kN/m	
phantom take, hinge lay	B2: carried, held 60 mm over the flap	no	none	0.002 m/s	0.17 kN/m	
phantom take, hinge lay	C0: laid (hinge), still held	no	carton 766	0.000 m/s	0.17 kN/m	
phantom take, hinge lay	C1: applied	no	carton 1715	0.000 m/s	0.19 kN/m	wet-out 95.1 %
phantom take, hinge lay	C2: after 2 s hold	no	carton 1715	0.000 m/s	0.19 kN/m	wet-out 95.1 %
phantom take, hinge lay	C3: after rub + settle	no	carton 1715	0.000 m/s	0.19 kN/m	wet-out 95.1 %
phantom take, hinge lay	**final**					**FAIL**: fails edge_band_wet_out>=90%_of_standard_application, no_pocket_over_20mm2, no_pocket_over_3mm2_in_barcode; reader: read
phantom take, arc lay	A': phantom hand took the label off the printer	no	none	0.004 m/s	0.06 kN/m	
phantom take, arc lay	B2: carried, held 60 mm over the flap	no	none	0.002 m/s	0.17 kN/m	
phantom take, arc lay	C0: laid (arc), still held	no	carton 929	0.000 m/s	1.26 kN/m	
phantom take, arc lay	C1: applied	yes (2)	carton 1214	0.039 m/s	6.97 kN/m	wet-out 54.2 %
phantom take, arc lay	C2: after 2 s hold	yes (2)	carton 1214	0.039 m/s	6.97 kN/m	wet-out 54.2 %
phantom take, arc lay	C3: after rub + settle	yes (2)	carton 1214	0.039 m/s	6.97 kN/m	wet-out 54.2 %
phantom take, arc lay	**final**					**FAIL**: fails wet_out>=90%_of_standard_application, edge_band_wet_out>=90%_of_standard_application, no_lifted_edge_segment, no_pocket_over_20mm2, label_not_torn, optical_decode_matches_payload; reader: no read
robot take + phantom handoff, wipe-on	A: after robot take	no	finger pad 64 35	0.002 m/s	0.19 kN/m	
robot take + phantom handoff, wipe-on	B1: robot hand released and retracted	no	none	0.261 m/s	0.13 kN/m	
robot take + phantom handoff, wipe-on	B2: carried, held 60 mm over the flap	no	none	0.000 m/s	0.13 kN/m	

Run	Phase	Torn (failed tri.)	Bonds	Max node speed	Peak membrane	Evaluator
robot take + phantom handoff, wipe-on	C0: wiped on (roller pass done, hand released)	no	carton 1797	0.001 m/s	0.30 kN/m	
robot take + phantom handoff, wipe-on	C1: applied	no	carton 1797	0.000 m/s	0.30 kN/m	wet-out 97.7 %
robot take + phantom handoff, wipe-on	C2: after 2 s hold	no	carton 1797	0.000 m/s	0.30 kN/m	wet-out 97.7 %
robot take + phantom handoff, wipe-on	C3: after rub + settle	no	carton 1797	0.000 m/s	0.30 kN/m	wet-out 97.8 %
robot take + phantom handoff, wipe-on	**final**					**FAIL**: fails no_pocket_over_20mm2; reader: read

- **The wipe-on lay passes the evaluator end to end** (roller at 20 mm/s, force-controlled through contact), in every run: the evidence run and three instrumented repeats (all 99.2 %). The paper label is taken off the printer, carried, wiped on and judged after the 2 s hold and the 5 N rub.
- Wet-out 99.2 %, edge band 94.9 %, largest pocket 6 mm², none in the barcode.
- No tear; carton moved 1.2 mm.
- The barcode was read at a declared pose.
- The label solver continued 896 of its 12,620 steps beyond 8 Newton iterations; 23 steps stayed above 10 µm (largest last update 0.17 mm).
- **The hinge lay is borderline.** It fails here (wet-out 95.1 %, edge band 85.2 %, a 71 mm² pocket and 6 mm² of trapped air in the barcode). A few square millimetres of trapped air decide it, so it is not a qualified technique; the wipe-on lay passes with margin in every run.
- **The arc lay** tears while laying (first failed triangles at 13.97 s, 5.7 kN/m) and pushes the carton 10.9 mm. The episode ends there (§14). It fails wet-out, edge band, lifted edge, pockets, crease and tear.
- **Why the roller starts on the tacked edge row.** A roller starting 6 mm beyond the edge seals a 24 mm² sliver behind the tacked edge row; presenting the edge 2 mm up so it meets the carton in the nip leaves the edge band at 84 %; and a 15° tack seals a 280 mm² strip.
- **The robot-to-helper handoff works; the chain after it is not qualified.**
- **The handoff.** In four runs (the rendered run above and three instrumented repeats) the handoff is clean.
- The fingertip patch releases completely under the helper's 90° peel: 35 → 0 nodes by 23.2–24.9 s. In the two repeats with a pad trace the membrane force stays ≤ 3.2 kN/m while the patch lets go.
- The hand withdraws without touching the label, in 2–8 steps to 33–39 mm clear.
- The carry is untorn.
- The handoff also holds at both pad-adhesion bounds and both coupled corners (§9.1).
- **The application.** In five runs (the rendered run above, three repeats instrumented step by step and one that followed every vertex's bond history) the handoff is clean in every run, and the wipe-on applies the label untorn: wet-out 95.5–98.1 %, the edge band passes (92.5–93.8 %), nothing in the barcode, and the barcode reads in the evidence run. The label stays untorn through the 2 s hold and the 5 N rub: after the wipe-on its fastest node stays at or below 24 mm/s and its peak membrane force at or below 1.13 kN/m, and in the three step-by-step repeats no pocket reads above 1 MPa at any step (at most 241 kPa). Every run still fails one check, trapped air over 20 mm² (largest pocket 53.6, 142.9, 107.2, 71.4 and 83.3 mm²), where the helper-only transfer leaves 6 mm². Some of these stand 13–104 µm off the carton, others lie in contact; in the run that followed every vertex, both pockets over 20 mm² are carton-bonded vertices released in the first 0.25 s of the wipe-on, while the hand held the sheet up at the tack angle, and barred from re-bonding under the roller (§15 item 16). They sit near the label's leading end.
- **Why the trapped-air rules of §4 matter here.** With a displacement-only convergence test, thin pockets squeezed by the roller's lift-off or the carton's spring-back stay pinned at the 0.1 µm gap floor while their pressure is orders of magnitude out of balance, shake the carton and release their stored gas energy as a kinetic kick. Replayed from checkpoints of the chain's own state, 9 of 10 such replays pinned pockets and kicked the label (up to 0.26 m/s and 4.1 kN/m, past the paper's 2.5 kN/m cross-direction strength), and one chain run tore under the 5 N rub (44.16 s, 7.0 kN/m). With the rules of §4 the replays from two further chain states never shook the carton (no step moving it more than 10 µm; its largest reaction 6.3 N, against 131–14,835 N without them) and never kicked the label (kinetic energy ≤ 2.7e-8 J), and the five chain runs

above stay untorn.

- **Why the choreography is as described.** A retreat toward the label pushes the pad face into the label edge and tears it, so the hand withdraws along the clearest direction. A still helper row lets the opening finger crease the label 90°, and the crease tears under the roller or leaves a lifted strip (wet-out 87–89 %), so the helper follows the pad. Following the opening finger takes the label onto the printer's peel bar, so the robot's hand moves opposite to that finger's opening. A peel that follows an assumed front speed kinks the label at the helper's row (74–77°), so the helper follows the measured front. The fold at the helper's row through the carry is 7.5°, as in the helper-only transfer (7°).

11. Robot demonstrations (unassisted)

Scope: what the scripted arm does with only the normal robot controls, on each robot configuration. One scripted attempt each; failures are evidence.

Taking the label from the printer: the arm nearest the printer approaches beyond the label's free end, fingers pointing back along it and closing along its normal (a card-edge grasp), grasps in force mode at 20 N per finger (Franka Hand semantics, §5), lifts steeply in 15 mm steps (the label peels off the presenter lip at a high angle), then draws up and away from the printer so the last row leaves the liner.

Preset / robot	Off the printer	Nodes on fingertip pads	Torn / failed triangles	Peak label speed	Velocity reflexes	wall/sim
parcel 4 x 6 paper	yes	35	no / 0	1.12 m/s	0	102
specimen 2 x 1 paper	yes	193	no / 0	0.88 m/s	0	48
part 2 x 4 PET	yes	51	no / 0	1.07 m/s	0	68*
parcel, dual-panda	yes	35	no / 0	1.01 m/s	0	87
specimen, dual-panda	yes	197	no / 0	0.75 m/s	0	78*
parcel, single-fr3	yes	35	no / 0	0.89 m/s	0	89
specimen, single-fr3	yes	193	no / 0	0.88 m/s	0	42

* Measured with 12 simulations running at once; not comparable with the rest of the column.

The take is qualified on all three configurations: the paper labels (parcel, specimen) and the PET label (part) come off the printer **without a tear**, with nothing left on the printer, on dual-FR3, dual-Panda and single-FR3. Repeated takes: the parcel take on dual-FR3 was clean in 4 of 4 instrumented runs and in every take of the evidence run (with the predicted-position contact rule of §5 it tears in 4 of 18). wall/sim: the numbers above were taken with other runs sharing the CPU quota; §13 has the dedicated measurement. Peak label speed: the fastest node moves 0.7–1.4 m/s for one or two 10 ms records while the peel front crosses the presenter lip. The paper springs off the lip as the bent strip's stored energy is released. This is the discrete front's snap, the same mechanism as the peel sawtooth (§15 item 1). It is not a reaction spike: no velocity reflex and no tear.

Hold and carry (unmodified take, 5 s still hold, 150 mm slide, the carry's fingers-down turn, 2 s hold): the force-mode grasp holds 19.9 N of pad-to-pad pinch throughout (20 N commanded); the held label settles to 0.14–0.30 mm/s net after each move; neither label tears. Peak membrane force in the fingers-down turn: parcel 0.47 kN/m; at rest the held parcel label carries 0.19 kN/m at the pad rim and 2 N/m beyond 12 mm (§15 item 19).

Carry, lay and press (one scripted attempt): **fails.** The take is clean (35 nodes on the fingertip pad) and the fingers-down carry reaches the carton without a tear. The sequence is: lay the label from its far end while still pinching it (225 nodes reach the carton); press the laid part with the other hand, stopping 40 mm short of the fingertip; open the fingers at 5 mm/s; back the fingertip off; press the whole label. The label tears at 13.17 s, in the first stroke of the laid-part press: the sliding fingertip drags the carton (79 mm by the end) while the other hand still holds the label's edge. Score 0.5; the partial score is a diagnostic only. The script's choices, each traced on the operator side and never in the world:

- the lay tracks straight Cartesian waypoints (a joint-space waypoint whips the pinched label through a ~1.3 m arc and tears it);
- the hand opens at 5 mm/s (at full speed it peels the pad patch at ~85 mm/s and tears it);
- the laid part is pressed first (releasing before any press leaves the label hanging from the pad);
- the press is 1.5 mm deep (4 mm drags the carton 10.8 mm in one stroke), and strokes out of reach are cut short (a joint-space approach to one sweeps the arm through the label).

What is left is the script's arm control: the press is position-servoed (depth sets the force; no Cartesian force control in this script) and the lay's last waypoint is out of the left arm's reach (16.7 mm, 14.8°), so the label is laid from a hand that cannot reach the near end. The assisted chain in §10.1 does these steps with force-controlled tools.

Fingertip pressing, assisted start: a labelled fixture drops the label 3 mm above the carton's rear flap (it sticks by tack); the right FR3 then presses it with closed fingertips in a grid of dabs (normal controls only). Result: no tear, the label bonded only to the carton, but wet-out reaches 34.5 % of the standard application: dabbing with two fingertips does not sustain a roller's pressure over a kraft face (wet-out follows sustained pressure, §4). It fails wet-out.

12. Stability under rest and legal abuse

Scope: stationary scenes stay stationary; legal actions cannot produce NaN, infinities or unexplained motion.

Rest (10 s simulated, no actions; render states every 0.1 s, rendered to fixed-camera videos):

Preset	Label	Items	Robot joints	Bonds	Label solver
parcel	5.75 mm in the first 2 s (the presented label finishing its droop), then at rest	carton 0.000 mm	< 1 µrad	630 → 630	no step beyond 8 iterations
specimen	0	tube 0.055 mm	2.0 mrad (the tube's free joint)	245 → 245	same
part / steel	0	0.000 mm	< 1 µrad	315 → 315	same
part / HDPE	0	0.002 mm	2 µrad	315 → 315	same

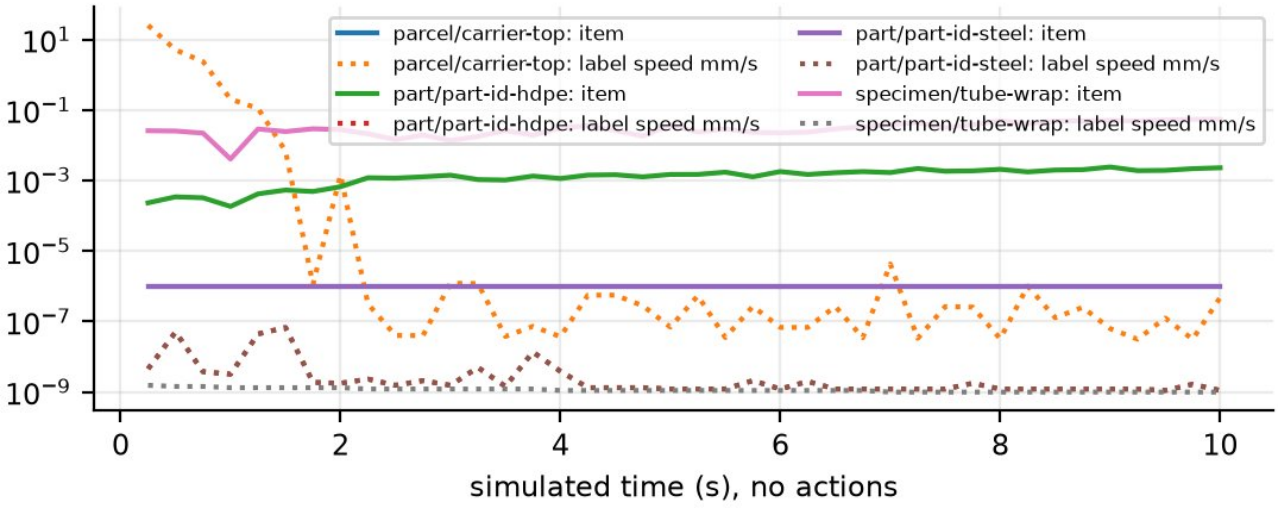


Figure 6. Item drift and label speed over 10 s at rest.

Legal-action stress (run twice with the same seed). Every random case starts from a **fresh presented label** (otherwise one label torn in the first case would leave every later case running on the frozen sheet): ten cases of random joint targets inside the limits for both arms, grip toggled at random. Then five directed cases act on a **label held by the robot** after an unmodified take: hold still, a 35 N grasp with a whip, dragging the held label into the carton, the grip toggled every tick, and the hand slammed into the bench. Speeds are reported per named body.

	run 1	same-seed repeat
all states finite, no solver fault	yes	yes
max rigid linear / angular speed	7.0 m/s / 13.2 rad/s (a fingertip, random targets)	same
velocity reflexes (joint > 1.05x its limit, arm stopped)	27	27
random cases that tore the presented label	1 of 10 (a flailing fingertip swept through it: 2 triangles, 3.6 kN/m)	same
directed cases that tore the held label	1 of 5: dragging it into the carton (1 triangle at 4.4 kN/m; repeat 2 at 5.5 kN/m)	same case
held label, hold / whip / grip toggling / slam	intact; peak membrane 0.17–0.19 kN/m	same
max fingertip pinch load	2.8 kN in one random case (the two pads slammed shut under a full-range random finger command); 20–35 N in every directed case	same
reactions reaching the tear-load cap	0	0
label steps not converged within 8 iterations	102 of 5,333	80 of 5,286

The repeat matches in every verdict (which cases tear, which stay intact, reflexes, speeds); the continuous quantities after contact differ slightly (e.g. the drag-into-carton tear: 1 vs 2 triangles, 4.4 vs 5.5 kN/m), because runs are not bit-reproducible (§15 item 7). Dragging a label that is bonded to the fingertip into the carton tears it, as it would a real label caught between a finger and a box. The linear speeds of up to 7 m/s come from joint-space commands inside the joint velocity limits; the FR3's 2 m/s Cartesian limit is not modelled (§15).

13. Runtime

Scope: wall-clock cost per simulated second (limit 200x), measured end to end through the public GAR API with the RGB observation returned and the action trace recorded, on a 16-CPU quota and one RTX 5090 shared with other sessions.

Render workers: one per operation (§14). Every RGB observation, optical read and diagnostic render runs in its own render-only Isaac worker, released as soon as that operation ends, including on errors; none is kept between operations. For an action the worker is started while the action's physics runs (warmed with one frame of the pre-action state), used for the observation and released right after capture on a background thread. What one such worker costs (parcel bench, 1280 x 720 `bench_rgb`, 24 samples per pixel):

Repeat	Scene export	Worker start to ready	First frame	Second frame (same worker)	Release	Total
1	3.9 s	6.9 s	3.3 s	0.13 s	1.9 s	16.1 s
2	3.7 s	6.5 s	3.4 s	0.13 s	1.9 s	15.5 s
3	3.8 s	6.8 s	3.3 s	0.12 s	1.8 s	15.8 s

The static stage is exported once per world and reused. A worker is ready 6.5–6.9 s after it is started (process, Isaac Kit, stage load: the shared worker's start, not this environment's code), its first frame takes 3.3 s (shader and scene warm-up) and every later frame 0.13 s. Warming the worker during the physics hides that first frame whenever the physics is longer than the start; it is never slower than rendering the first frame at the observation. What remains visible to a policy is max(0, ~10.1 s − physics) + the frame.

Light actions: `gar.create(...)`, then `reset()`, then six 0.5 s `NamedJointTarget` actions (250 label ticks each), each traced like the HTTP server's and returning the decoded 1280 x 720 `bench_rgb` image. Every cycle checks the simulated time three ways (world ticks, the observation's `sim_time_s`, the request) and records where its wall time went: physics (rigid step, label solver, coupling), the wait for its early-started worker, the frame, and the rest (state, trace, image write). Startup (create + reset, including the first worker) is reported separately.

Preset / robot	Startup	Mean wall/sim, 0.5 s traced action + RGB	Worst action	Per action: physics / render wait / frame / rest (mean)	Clock check	RGB decoded
parcel / dual-fr3	56 s	40x	45x	19.3 / 0.0 / 0.72 / 0.11 s	0.5 s = 0.5 s	True
parcel / dual-panda	55 s	36x	42x	17.3 / 0.0 / 0.72 / 0.03 s	0.5 s = 0.5 s	True
parcel / single-fr3	54 s	41x	47x	19.6 / 0.0 / 0.70 / 0.13 s	0.5 s = 0.5 s	True
part / dual-fr3	37 s	24x	24x	11.3 / 0.0 / 0.68 / 0.04 s	0.5 s = 0.5 s	True
part / dual-panda	36 s	22x	23x	10.4 / 0.1 / 0.68 / 0.05 s	0.5 s = 0.5 s	True
part / single-fr3	34 s	22x	22x	10.2 / 0.0 / 0.66 / 0.03 s	0.5 s = 0.5 s	True
specimen / dual-fr3	41 s	25x	26x	12.0 / 0.0 / 0.66 / 0.05 s	0.5 s = 0.5 s	True
specimen / dual-panda	40 s	23x	23x	10.8 / 0.1 / 0.65 / 0.03 s	0.5 s = 0.5 s	True
specimen / single-fr3	38 s	23x	23x	10.9 / 0.0 / 0.64 / 0.03 s	0.5 s = 0.5 s	True

- **All nine combinations stay under the 200× limit:** 22–41× mean, 47× worst single action. Each 0.5 s action's 10–20 s of physics hides the worker's start completely (render wait ≤ 0.1 s on average), and the frame costs 0.64–0.72 s. A worker kept for the session gives 21–38×, and a new worker started at the observation 52–71×. Each action advanced exactly 0.5 s on all three clocks and every image decoded to 1280 × 720. Startup takes 34–56 s; the parcel bench, with the largest label, is slowest.
- The runs went alone among this environment's processes; other long-running sessions shared the machine; every action records the host's 1-minute load (3.4–35).

Active contact, 10 Hz: a scripted command stream is recorded and replayed through `gar.create` in 0.1 s actions, each traced and returning the RGB image. Two workloads: the parcel take (pinch, steep peel off the presenter lip, draw-away; 6.5 s, must come off the printer untorn) on the three robots, and the **complete carrier-top job** of the scripted demo on dual-FR3 (take, carry, lay, press of the laid part, fingertip off, full press with the other arm, settle). Every action is kept in the record with its split; the five slowest of each workload are listed below.

Workload / robot	Action length	Actions (traced, RGB)	Overall wall/sim	Median / p90 / worst action	Per action: physics / render wait / frame / rest (mean)	Replayed result	Label steps capped / extended / unresolved
parcel take / dual-fr3	0.1 s	65	132x	114x / 180x / 264x	9.3 / 3.1 / 0.71 / 0.07 s	taken, 35 on pads	15 / 775 / 1 of 3450
parcel take / dual-panda	0.1 s	65	133x	114x / 188x / 210x	9.2 / 3.3 / 0.72 / 0.07 s	taken, 35 on pads	20 / 715 / 4 of 3450
parcel take / single-fr3	0.1 s	65	133x	115x / 179x / 278x	9.5 / 3.1 / 0.69 / 0.05 s	taken, 35 on pads	13 / 788 / 0 of 3450
complete carrier-top job / dual-fr3	0.1 s	608	144x	112x / 216x / 458x	11.8 / 1.8 / 0.72 / 0.05 s	taken, 42 on pads, 0 on the carton	386 / 8918 / 97 of 30596

Slowest actions (all actions are in the records):

Workload / robot	Action #	Kind	Sim	Wall	wall/sim	Physics	Render wait	Frame	Label steps capped / extended (cumulative)
take / dual-fr3	50	joints	0.10 s	26.4 s	264x	25.7 s	0.0 s	0.73 s	9 / 503
take / dual-fr3	36	joints	0.10 s	20.2 s	202x	19.4 s	0.0 s	0.72 s	0 / 73
take / dual-fr3	55	joints	0.10 s	18.7 s	187x	18.0 s	0.0 s	0.71 s	10 / 594
take / dual-fr3	37	joints	0.10 s	18.7 s	187x	17.9 s	0.0 s	0.72 s	0 / 121
take / dual-fr3	42	joints	0.10 s	18.3 s	183x	17.6 s	0.0 s	0.71 s	0 / 282
take / dual-panda	52	joints	0.10 s	21.0 s	210x	20.3 s	0.0 s	0.72 s	15 / 532
take / dual-panda	41	joints	0.10 s	20.1 s	201x	19.3 s	0.0 s	0.72 s	7 / 234
take / dual-panda	50	joints	0.10 s	19.5 s	195x	18.5 s	0.0 s	0.72 s	12 / 478
take / dual-panda	42	joints	0.10 s	19.4 s	194x	18.7 s	0.0 s	0.72 s	7 / 283
take / dual-panda	49	joints	0.10 s	19.0 s	190x	18.0 s	0.0 s	0.71 s	11 / 451
take / single-fr3	50	joints	0.10 s	27.8 s	278x	27.0 s	0.0 s	0.70 s	6 / 499
take / single-fr3	60	joints	0.10 s	19.3 s	193x	18.6 s	0.0 s	0.70 s	13 / 767
take / single-fr3	44	joints	0.10 s	19.3 s	193x	18.6 s	0.0 s	0.69 s	0 / 346
take / single-fr3	42	joints	0.10 s	19.0 s	190x	18.3 s	0.0 s	0.69 s	0 / 281
take / single-fr3	36	joints	0.10 s	18.8 s	188x	18.1 s	0.0 s	0.70 s	0 / 73
complete job / dual-fr3	434	joints	0.10 s	45.8 s	458x	45.0 s	0.0 s	0.73 s	357 / 3810
complete job / dual-fr3	80	joints	0.10 s	45.3 s	453x	44.5 s	0.0 s	0.73 s	101 / 1506
complete job / dual-fr3	81	joints	0.10 s	43.1 s	431x	42.4 s	0.0 s	0.74 s	120 / 1556
complete job / dual-fr3	77	joints	0.10 s	43.1 s	431x	42.3 s	0.0 s	0.72 s	45 / 1356
complete job / dual-fr3	79	joints	0.10 s	42.6 s	426x	41.5 s	0.0 s	0.73 s	67 / 1456

- **The per-operation rule costs about a third at 10 Hz.** The take averages 132–133× over the whole take on the three robots, median 114–115×, 90th percentile 179–188×; with a worker kept for the session it is 99–104× (median 79–87×). A 0.1 s action's physics (median ~8 s) is shorter than the ~10.1 s worker start, so the observation waits 3.1–3.3 s on average for its worker, and the worker's start competes with the physics for the CPU quota.
- **The worst actions are physics, not rendering:** the slowest 0.1 s action of each robot costs 210–278×, and each of the five slowest per robot has 17.6–27.0 s of physics, no render wait and a 0.7 s frame — the label solver's adaptive Newton continuation at the peel front (8 → up to 32 iterations). The replayed take comes off the printer untorn on all three robots.
- **The complete carrier-top job** (608 actions, 60.8 s simulated, 2.4 h of wall time) averages 144×, median 112×, 90th percentile 216×; 84 of its 608 actions (14 %) exceed 200×, the slowest 458×. Its slowest actions are all physics (41.5–45.0 s each, no render wait): the carry's fingers-down turn (7–8 s into the job) and a press stroke (43 s). Its render wait averages 1.8 s (its physics per action is longer than the take's). Every image decoded. The replay is open loop: it did not reproduce the scripted run (the replayed label ended on the fingertip pad, 42 nodes, none on the carton, untorn) — this row measures the workload, not a task result (§11 has the scripted run itself).

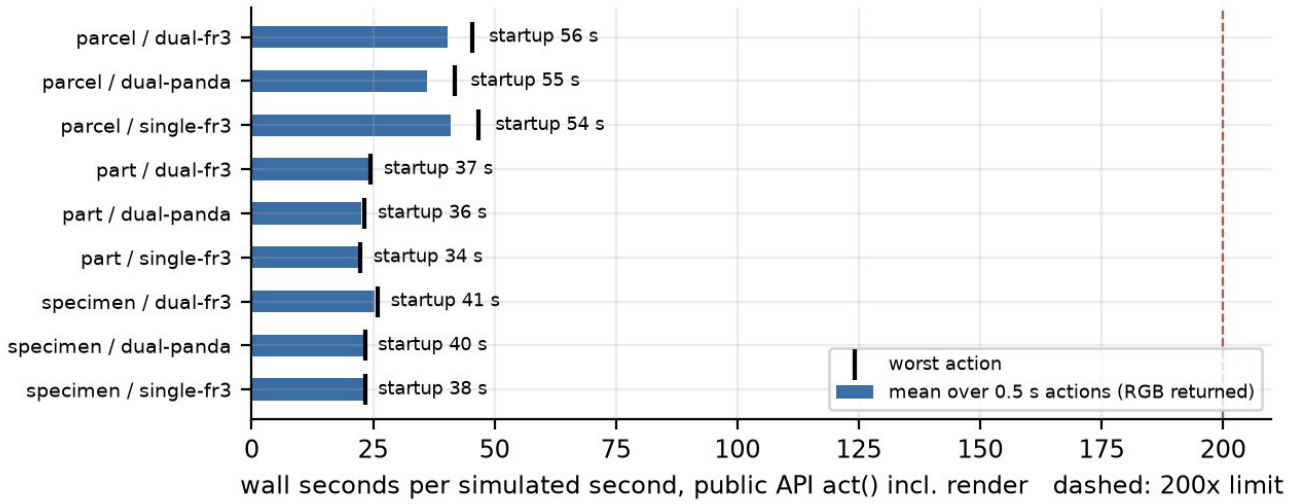


Figure 7. End-to-end wall/sim per 0.5 s action with the RGB observation, per preset and robot; dashed: the 200x limit.

The runs in §10–§12 report their own wall/sim; most of them shared the quota with up to ~15 other simulations during the evidence batches and are not runtime measurements.

The trapped-air rules of §4 and these timings. The timings above were measured without the trapped-air rules of §4, which act only on label steps with an enclosed gas pocket. Counted with them: the parcel take and the specimen take have none in 3650 label steps each, so their timings are unchanged; the light 0.5 s actions swing one arm's first and fourth joints by ±0.05 rad and never touch the presented label, which holds no pocket until a finger closes on it (the part take's first forms at its pinch, 3.11 s); the PET part take has one in 946 of its 3650 steps and needs 1.6 % more Newton iterations with the rules (18,783 against 18,495, the same take); the scripted carrier-top job's take, lay and press (13.6 s simulated, the first part of the 60.8 s complete-job replay) has one in 71 of its 6796 steps, at most +0.61 kPa; the rest of that replay was not counted. In the robot-to-helper chain's settle, hold and rub, replayed from identical states, the pocket convergence test alone takes 15–75 % more Newton iterations than none of the rules, and all six rules −31 % to +2 %.

14. Interfaces

Scope: how a policy, an operator and a reviewer use the environment.

- **Launch:** `gar run label-application --select preset=parcel --select task=carrier-top --select seed=0 --policy-key` (selectors or a `--task-file`; `gar tasks label-application` lists the six entries).
- **Robots** (shared catalog, `robots/rigs/label-*.json`): `dual-fr3` (default), `dual-panda`, `single-fr3`. The single FR3 stands on the left pedestal beside the printer (the right pedestal is 0.95 m from the printer exit, beyond the FR3's 0.855 m reach). The Panda URDF models its fingers as collision meshes; the same Franka Hand fingertip boxes as on the FR3 are authored for label contact on them and the finger meshes carry the pinch load. Each configuration is qualified by a real take (§11), not only by build-and-step. G1/humanoid setups from the catalog are **not supported** here, and the catalog says why (`robots/catalog.py unsupported`): `robot="g1"` fails at launch with that reason, not a generic "unknown setup". The environment reads every per-hardware fact from the catalog (§5), so supporting a new arm or hand is catalog work (a `contact.fingertip_pads` spec, a gripper group, a ready pose) plus label-contact qualification. For the G1 that means a pinch-grasp model for the three-finger Dex3 hand, arm IK on the whole-body chain, bench cameras and a qualified take; it is not in scope.
- **Policy actions (restricted key):** `named_joint_target` (rad / m), `gripper` (`half_width_m` or force-mode `inward_effort_n` ≤ 70 N), `cartesian_pose` / `cartesian_delta` (IK to joint targets). One public tick is one label step, 2 ms (= 2 rigid 1 ms ticks); `NamedJointTarget.ticks` counts label steps and duration-based actions must be multiples of 2 ms.
- **Episode end:** a torn label ends the episode: `terminated` is true in the step result for owner and policy access, `info.termination` records the tick, time and failed-triangle count, and the label is frozen where it tore and loads nothing any more (`config.stop_on_tear`, default on). The tear onset is the validated quantity; what an element-eroded sheet does afterwards is not (§15), and the evaluator's `label_not_torn` can never pass again, so continuing would only simulate artefacts.
- **Solver quality in every step:** `info.label_solver` (and `state.label_solver`) reports the label solver's steps, steps not converged within the base 8 Newton iterations, steps continued beyond them (adaptive, up to 32), steps still moving by more than 10 µm after that (`unresolved`) and the largest such last update.
- **Policy observations:** joint positions and velocities, RGB from the overhead `bench_rgb` and the mounted wrist cameras (single-fr3: `left_wrist_rgb` only; requesting a camera the rig does not mount is rejected), the instruction. No label geometry, bonds, poses or evaluator internals.
- **Privileged (owner key):** full state (label summary, bonds by surface, pockets, pad loads, item poses), label mesh, evaluation, per-command SQLite trace, checkpoints, diagnostic cameras.
- **Checkpoints:** `save_checkpoint` / `list_checkpoints` / `restore_checkpoint` follow the GAR backend contract: restore returns `(state, info)` with the checkpoint label, tick, generation and continuity `restored_discontinuity`. The label checkpoint includes bond, damage, gas, pockets, erosion, pinch contact history and re-bond state.
- **Dependencies**: python-barcode (the label symbols, drawn at creation), zxing-cpp (the verifier's decode) and pypardiso (the label's linear solves) are declared as the extra `gym-anything-robotics[label-application]`. `gar doctor label-application` checks the three imports, that pypardiso loads MKL (`PYPARDISO_MKL_RT` is needed only when the interpreter's prefix has no MKL, e.g. a venv layered over the runtime), and reports `ISAAC_RENDER_PYTHON` (RGB and reader views; the environment runs without cameras).
- **Tests:** contract tests (CPU, seconds): the six tasks resolve, unknown config fields fail, the catalog offers the three rigs with the environment's own controller profile, G1 and unmounted cameras are rejected with reasons, the printed symbols decode to their payloads, doctor declares the dependencies, a force-controlled tool presses through contact while a hovering one does not, wet-out follows sustained pressure (a fast pass wets less than a slow one; a 2 ms tap stays under a tenth of the instantaneous law), and the liner web blocks unbonded points. With `GAR_LABEL_PHYSICS_TESTS=1` (GPU, minutes): restore returns `(state, info)`, the policy view is redacted, and a force-mode grasp holds its command. With `GAR_LABEL_RENDER_TESTS=1` (GPU, Isaac, minutes): no render worker survives a completed reset, action or evaluation or a failed action (below). All 14 pass (9 fast + 3 physics + 2 render lifecycle; 130 s).
- **Session output:** auto-created sessions go to `$GAR_WORKSPACE/.gar/label-application-sessions/` (the workspace's git-ignored managed session area) (else the system temp directory), never into the package or `site-packages`; the 20 newest are kept, and only folders carrying the backend's marker are ever pruned.
- **RGB render worker:** one per operation (render-only workers are released when their operation ends, including on errors). An action starts its worker while its physics runs, warmed on a snapshot of the pre-action state; the action's observation renders with it and releases it right after capture, off the reply's path. An evaluation's optical read and every diagnostic render do the same; a failed action releases its worker before the error returns. No worker is kept between operations. The static stage is exported once per world and reused; every frame sends the live body poses and label mesh. Session state, the action history and later renders are unaffected (a render after a completed operation starts a fresh worker). Tested with `GAR_LABEL_RENDER_TESTS=1` (`LabelRenderLifecycleTest`: no worker survives a completed reset, action or

evaluation, or a failed action; each action starts exactly one; the action after the reset renders with a fresh worker). Cost: §13.

- **Packaging:** verified from an installed wheel in a clean virtual environment: all three robots create, reset, act, save/list/restore (tick back to the saved value) and decode the environment's printed symbol with the declared dependencies; single-FR3 rejects the right wrist camera. External prerequisites, not in the wheel: numpy/scipy/newton/warp, `PYPARDISO_MKL_RT` pointing at `libmkl_rt.so.3`, the Panda description (`TITRATE_PANDA_URDF`, shared robot catalog asset) and, for rendering, Isaac Sim via `ISAAC_RENDER_PYTHON`.
- **Live HTTP transaction**: a real `gar run` session; the policy key reads capabilities, task and observation and commits an action, and is refused checkpoint and state routes (403); the owner key saves a checkpoint, acts, restores and reads state and evaluation; the session is stopped. Measured: the policy action (50 ticks) advanced the observation clock 0.0 → 0.1 s; the owner saved at tick 50, acted to tick 150, restored, and state read tick 50 again (`restored_discontinuity`); the evaluation returned success false, score 0.4 (nothing applied). Wall times (per-operation render workers): 11.3 s for the first observation, 11.8–12.2 s per 0.1 s action, 10.5 s for the restore (re-render), 12.3 s for the evaluation (reader renders); the policy's refused requests and the owner's state reads take ≤ 4 ms. The keys are read from the launcher's output in memory only and are not recorded.
- **End-to-end timing** through the public API with the RGB observation returned and the action trace recorded, light and active-contact workloads: §13.

15. Limitations

Scope: what the world does not do. Read before relying on it.

1. **Peel discretisation.** A 3 mm node row releases at once; the mean peel force needs a numerical factor per mesh size, fitted at 90° to one datasheet point (c_{90}) and at 180° to the elastic Kendall solution (c_{180}, numerics only), with the factor interpolated in $(1 - \cos\theta)$ in between; the force trace is a sawtooth whose peaks are mesh artefacts. On the 3 mm mesh the steady 180° peel is **bistable** in the factor (0.73 → 243, 0.735 → 147, 0.74 → 291 N/m): no factor puts it on the elastic 190 N/m. With the calibrated factor the 3 mm mesh reads 143.1 N/m in §7.1 and 158.1 N/m in the convergence run of §8, 17–25 % below it. 2 mm and 1.5 mm stay on the elastic answer (189.3 and 190.8 N/m against 190).
2. **The adhesive law is elastic in angle; the datasheet's angle dependence is bounded, not applied.** 3M 7818's 90°/180° pairs imply $\Gamma_{180}/\Gamma_{90} = 2.2–3.4$ (backing and process-zone dissipation). The peel angle a 3 mm front node can read is too coarse to carry an angle law: a peel at exactly 90° reads a median 69° (49–96° after smoothing), one at exactly 180° reads 119° (76–152°), and a law applied to that reading raises the calibrated 90° point by 24 %. Where it matters: the liner and silicone exit plate are referenced to a 180° datasheet release, so the elastic 180° peel reproduces them; applications peel at low angle; the fingertip-pad bond (open proxy, 7818 LDPE 130 J/m² at 90°, 440 at 180°) carries the gap and is bounded 2–440 J/m² in §9.1.
3. **No interfacial slip on silicone release surfaces.** PSA on PDMS slips at low shear stress in reality (literature, qualitative); here the liner and exit plate resist in-plane drag as a lap joint (Kendall), probably too strongly. No published shear-slip value was found to anchor a slip law, so none is guessed.
4. **Least-grounded constants:** roughness pressures, w_0, n, paper E, CD strength and yield strain, fingertip-pad adhesion and compliance are estimates; §9.1 bounds how they move the task judgments.
5. **Rate law at high speed:** 50 mm/s peel departs from the rate law (§7.1).
6. **Self-contact is frictionless and non-adhesive** (a label folded adhesive-to-adhesive does not stick to itself), and pairs closer than ~1.5 elements in the rest state are excluded (a crease tighter than the element size is not resolved).
7. **Label ↔ robot coupling is lagged (one label step), filtered (5 ms) and capped at the tear load.** The velocity reflex checks joint speeds only; the FR3's 2 m/s Cartesian speed limit is not modelled (random full-range joint targets reach 5–6 m/s at a fingertip, §12). Repeated runs are not bit-reproducible. **Newton convergence is adaptive and reported.** A label step that has not converged (last update ≥ 1 µm, or a trapped-air pocket's last volume update ≥ 0.1 %) after 8 Newton iterations continues up to 32; one still moving by more than 10 µm (0.1 label thickness) after that is counted as unresolved and reported in every step's info (§14). In the reported runs: the eleven assisted applications and controls never needed more than 8 iterations; the pressing demo continued 5 of 9,930 steps and resolved them all; the parcel and part takes continued 16–26 % of their 3,650 steps and left 0–9 unresolved (largest last update 0.06 mm); the four transfers (§10.1) continued 8–21 % and left 16–25 unresolved (≤ 0.27 mm). **The 2 x 1 specimen take is the least resolved run:** 81 unresolved steps on dual- and single-FR3, 107 on dual-Panda, with last updates up to 0.34 mm, over three label thicknesses. Its verdict (off the printer, no tear) therefore rests partly on steps the solver did not fully converge, and so does its sensitivity (§9.1: no setting tears it). A fixed 8-iteration cap would accept such steps silently: 19–35 % of take steps, last updates up to 0.47 mm.
8. **Squeeze set-back:** where fingers overlap each other or an item by up to ~1 mm, the label sits at the mid-plane; its thickness never stops the fingers (it is not in the rigid solver).
9. **Hour-scale flagging and creep are not simulated;** the post-application hold is 2 s.
10. **Nothing after a tear is simulated as paper.** The tear onset is validated (averaged orthotropic criterion, calibrated strengths) and is the only tear quantity any check uses. Element erosion after it breaks a still-loaded sheet into element-sized pieces, which real paper does not do (this "confetti" can drive a carton 132 mm). A tear therefore ends the episode: the label is frozen where it tore and transmits no load (§14), so no debris acts on robots or items.
11. **No task is solved by a policy or an unassisted script end to end.** Robot takes are unassisted (§11). The printer-to-carton chain passes the evaluator with a phantom-hand take, carry and wipe-on lay (§10.1). The robot-to-helper handoff works, but the chain after it is not qualified: it traps more than 20 mm² of air near the former fingertip patch (§10.1, item 16).
12. **The HDPE light-load control passes:** smooth surfaces (p_r = 2 kPa, an estimate) wet out at 1/10 load (98.9 %), so the part-id-hdpe task does not reject a light touch. At p_r = 20 kPa it does (42 %) while the full-load application still passes (99 %, §9.1); it also fails at 3x the wet-out time constant (87 %). No measurement anchors p_r on HDPE, so the baseline is kept and the gap is stated.
13. Printer housing, stand, lips and item dimensions are authored; the dispensing mechanism runs as a component test while the workcell starts from the presented label.
14. Visual dressing (room, props, printer shell, tube glass and blood) has no physics; cylindrical rigid colliders are 32-facet hulls.

15. **G1 is not supported** (rejected at launch with the reason: the G1/Dex3 hands have no label-contact fingertip pads, pinch-load mapping, arm IK or bench cameras qualified here); it is not in scope. zxing-cpp and python-barcode are declared (`label-application` extra); `gar doctor` checks them (§14).
16. **A small fingertip bond patch releases at 5 mm/s; fast patch peels tear.** With the patch balance (§4) the fingertip patch comes off a 50 mm label at 90° untorn at 5 and 50 mm/s, the straight pull at 5 mm/s, the 180° peel-back at 5 mm/s, and the robot-to-helper handoff's 90° peel at 5 mm/s (§7.5, §10.1). It still costs about twice Γ at 5 mm/s (260 J/m²; 483 J/m² at 180°, where the sub-element fold re-bonds). At 50 mm/s the straight pull and the 180° peel-back tear without releasing a node, and a fingertip opening at ~85 mm/s during the scripted lay tears the 101.6 mm parcel label at its pad patch. A real paper label pulled that fast off a rubber fingertip may well tear too (Γ(50 mm/s) = 517 J/m² needs ~9 kN/m at a shallow angle), but no measurement separates that from the discretisation: the demonstrations peel patches at 5 mm/s. A just-released row that falls back onto its substrate re-tacks and must release again. On the adopted law the pinch probe's re-tacked 13-node island does not, and the strip tears at 0.81 s (§7.5); the demonstrated robot takes do not reach this. A re-tacked island of 1–3 nodes loaded in its plane does not: it has no interior, and the strip reading returns ~0 there (at a 90° front the fold, ~1.5 mm, is shorter than the 3 mm element, so a released row stays against its substrate). Two extensions of the patch balance release it (Kendall's balance for interior-less islands, Kendall's 0° term for pressed patches loaded in-plane). They are off: with both on, the helper-only wipe-on passes and the pinch strip releases untorn, but adopting them changes the front law wherever an island has no interior or is pressed and pulled in its plane, and needs its own validation and sensitivity study. The handoff's re-tack is not this limit: it is the robot's retreat pushing its pad face into the label edge (3 of 3 instrumented repeats); with the clearance-seeking withdrawal nothing re-tacks, and the latent limit is not reached by any demonstration. The handoff chain's remaining failure is elsewhere (§10.1):

- **Trapped air.** Every run of the chain traps more than 20 mm² near the former fingertip patch: 53.6–142.9 mm² in five runs, where the helper-only transfer leaves 6 mm². Much of it is not air. A run that followed every vertex found both of its pockets over 20 mm² (83.3 mm² lying in contact, 23.8 mm² at 4–9 µm) made of vertices that were bonded to the carton and released from it in the first 0.25 s of the wipe-on, while the hand held the sheet up at the tack angle (the carton bond count rose to 188 and fell to 172). A vertex released while in contact may bond again only after it has separated from every surface (§4: a front under shear must not heal every step), so the roller that then pressed them could not re-bond them. Its only standing air bubble is one node, 60 µm high. The other runs also carry unbonded patches in contact at the gap floor, some at a vacuum (down to −68 kPa), and pockets standing 13–104 µm off the carton. **A pressure-sensitive adhesive pressed back by a roller re-bonds**, so this rule, not only the operator's tack, is part of the chain's failing check. Letting a released vertex re-bond under a compressive load changes the peel, pinch, lap and liner evidence and needs its own validation; it is open work.
- **The trapped-air model at the mesh's resolution.** No pocket in the three step-by-step chain runs reads above 1 MPa (at most 241 kPa). At mesh scale a pocket is at least one node (6.0 or 11.9 mm² on the 3 mm mesh) and two nodes are 17.9 mm², so the evaluator's 20 mm² limit is about two nodes; air gathers node by node as pocket vertices bond, so when several vertices bond in one step their air lands in one node at once. In replays that switch the rules of §4 on from a state with pinned pockets (air compressed ~100x), such a node read up to 305 MPa for one step and re-inflated at 0.11 m/s (0.53 kN/m), and once 35.6 MPa for one step under the rub finger. The same finger over a one-node pocket holding that much air on a static plate reads at most 154 kPa, with the reading and the solve agreeing; the 35.6 MPa step moved nothing (fastest node 0.048 m/s, 0.30 kN/m) and is not traced further. Pockets stay sealed on porous kraft (item 21), down to −101 kPa.
- **What is not established.** Whether the pockets near the former patch would form on a real label.

17. **Single contact actions at 10 Hz can exceed 200×, and one worker per operation costs about a third at 10 Hz.** With a render worker per operation (§13, §14) a contact-heavy take with an RGB image every 0.1 s averages 132–133× over the take on the three robots (median 114–115×); a worker kept for the session gives 99–104×. The worker's ~10 s start is only partly hidden behind a 0.1 s action's physics, and competes with it for the CPU quota. The hardest 0.1 s actions of the take cost 210–278×, on physics alone (the label solver's adaptive Newton continuation at the peel front), not rendering; replayed as a whole at 10 Hz, the complete scripted carrier-top job averages 144× and 14 % of its 608 actions exceed 200× (worst 458×, in the carry's fingers-down turn and a press stroke, again physics). Light 0.5 s actions hide the worker completely and run at 22–41×. Reducing the worker's own start is in the shared Isaac render worker, not in this environment.
18. **Wet-out kinetics are derived, not measured.** Wet-out follows the pressure sustained over $\tau_w = 0.16$ s (§4), anchored to the 1 Hz reference of the adhesive modulus that p_r rests on. No measured PSA contact-formation time series was found; §9.1 bounds τ_w ×⅓..×3, and **at 3x the kraft applications fail** (parcel 86 %, relabel edge band 88 %): the kraft verdicts rest on this estimate. Slow wet-out growth at rest (the Creton–Leibler creep that the datasheet's 10 min → 72 h dwell gain reflects) is carried by the dwell law, not by φ.
19. **Stress where a pinched sheet wraps the fingertip's rounded corner.** A held 2 x 1 specimen label carries up to 0.79 kN/m on the triangles where it wraps the pad's 2 mm corner fillet (a doubly curved region:

the pinched nodes there span 1.1 mm out of plane over ~3 mm); 0.03 kN/m median elsewhere; the parcel label 0.19 kN/m at the rim and 2 N/m median beyond 12 mm (its weight over its width is 0.15 N/m). Real paper would wrinkle there; the 1.5 mm mesh cannot, and carries the misfit as membrane stress. It is stationary (does not grow in a slide or the carry turn) and below the tear load.

20. **Dispensing presents forward through a bending release that is derived, not measured** (§7.2): the released paper leaves the peel edge and cantilevers forward (strip 2.0°, 4 × 6 label 12.4° with gravity droop, no curl-back; a limp control stays on the web), because a front on the curved web releases ½ D κ^2 capped at the facestock's plastic moment and edge nodes read the bending stored behind them. The bending stiffness and yield strain of the facestock are handbook-level estimates (§6), so the exit angle is a model output, not a calibrated observable; no measured dispensing trajectory was found to compare.
21. **Trapped air is sealed on every substrate, including porous corrugated kraft.** An enclosed pocket keeps its gas amount (§4), which is right for glass, steel, HDPE and the PET tube but not for kraft.

- **What the data say.** Kraftliner is air-permeable: Mondi's ProVantage Kraftliner datasheet gives Gurley ≤ 150 s (ISO 5636-5), i.e. an air permeance P ≥ 127/150 ≈ 0.85 µm/(Pa·s) (ISO 5636-5's conversion); testliner for bags is 5–60 s (2–25 µm/(Pa·s)). The air under a label on a carton can pass through the liner into the flutes, so a pocket of gap h relaxes its excess pressure in about $\tau \approx h/(p_{atm} P)$: ≈ 1.2 ms for h = 0.1 mm, less for thinner pockets, shorter than one 2 ms label step.
- **What the model does instead.** Pockets on kraft in the robot-to-helper chain read from −90 to −101 kPa (a sealed pocket re-expanding), where real air would stay near atmospheric. A pocket vertex pressed into contact bonds and passes its air to the rest of the pocket, so a blister that a roller presses flat gathers its air into a smaller bubble; with the air escaping through the board it would wet out completely.
- **What still holds.** The evaluator's pocket checks count enclosed unbonded area, which remains a defect a person would see, whatever its pressure.
- **Open work.** A per-substrate venting term is the modelling step. It would change the kraft tasks' trapped-air outcomes, including the robot-to-helper chain's remaining failure, and needs its own validation.

16. Reproduction

From the release's repository root, after installing it with the `label-application` extra as its README describes:

```
gar doctor label-application      # python-barcode, zxing-cpp, pypardiso and MKL, the Isaac render interpreter
gar tasks label-application       # the tasks and their selectors
gar run label-application --select preset=parcel --select task=carrier-top --select seed=0 --policy-key
gar verify label-application      # the task evaluation of the running session
```

`gar run label-application --robot dual-panda` (or `single-fr3`) picks another configuration (§14). Repeated runs are not bit-reproducible (§15 item 7).